Analyzing a Longitudinal Study of Cognitive Decline Using Mixed-Effect Models

by

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1 Introduction

1.1 Cognition in the Elderly

Declining cognition is an unfortunate reality for many people as they age. It can lead to dementia, where patients with dementia struggle to lead independent lives and are a risk to their safety and others. There are many factors believed to contribute to dementia: this research aims to find which factors lead to the steepest decline in cognition. Some factors may be due to the lifestyle of people, such as smoking or lack of exercise, and other factors cannot be changed, such as ethnic background [6]. It is important for people to be aware of lifestyle factors that impact cognition, to reduce their cognitive decline and to have their cognition tested sooner if they feel at risk, to lessen the effects of cognitive decline [5].

The Health and Retirement study provides data from a 30 year longitudinal study on US adults over 50. From the data, the factors contributing to the steepest decline in cognition as the respondents age can be investigated. This research uses weighted linear mixed-effects models to measure cognitive decline, where fixed effects coefficients are used to derive population level inferences. Previous cognition studies from the Health and Retirement study typically perform a survival analysis. One study on the HRS data using mixed-effect models measures the association of the big 5 personality traits with cognition, particularly that neuroticism has a high association with a lower cognition [8]. A potential flaw in their study is that neuroticism is highly associated with depression and depression is often identified as a factor contributing to cognitive decline in studies.

Section 1 will discuss general methodologies for longitudinal data and survey sampling. Section 2 will expand further on these methodologies and how they are applied to the HRS data. Section 3 will discuss the data source and how the weighting was handled. Section 4 will investigate whether a mixed-effect model is necessary for the data and explore potential covariates. Section 5 will apply the modeling methodologies to the data and interpret the results. Section 6 will draw conclusions from the models and elaborate on future work.

1.2 Longitudinal Data

Longitudinal data occur when repeated measures are collected on the same subjects over time, resulting in multiple observations per subject at various times. The benefit of a longitudinal analysis over a cross-sectional analysis is the addition of the cohort and aging effects. Analyzing the cohort effect requires observing the differences in individuals from different cohort groups (generation, for example), whereas the aging effect requires observing the changes in the response variable over time within individuals [10]. Both effects cannot be analyzed in cross-sectional studies. Longitudinal studies can also identify causality as associations of what causes a certain change in units. The causal inferences found in uncontrolled studies will not hold as much weight as the inferences found in controlled studies, but the associations can still lead to meaningful insights.

Longitudinal studies have disadvantages, one of which being the cost of conducting the study. There is a lot of planning and logistics to keep track of in a longitudinal study because individuals are sampled repeatedly. Generally, the cost of a longitudinal study that goes through n waves is more expensive than n individual cross sectional studies if the sample sizes are the same. An additional disadvantage in data collection in prospective longitudinal studies is panel conditioning. This occurs when the response from an individual in a later wave differs from what it would have been in their first wave. This can include a respondent learning new information through questions in a previous wave or a respondent feeling more comfortable being surveyed and revealing more sensitive information than they would during their first interview. Sample attrition is another problem in prospective longitudinal studies in which the respondent is not responsive [7].

Another factor that differentiates longitudinal studies from cross-sectional studies is that repeated measures on subjects result in a correlation between observations for the same subject. Models used in cross-sectional studies, such as ordinary least squares regression, require the assumption that all observations are independent, which is violated in longitudinal data due to the aforementioned within-person correlation. However, the observations between two different subjects are independent, resulting in a correlation of zero. The covariance matrix in longitudinal studies has block diagonal elements, where the rows and columns corresponding to the same subject are non-zero, while the rest are zero. If all observations are kept in the vector $\vec{y} = [y_{11}, y_{1,2}, \ldots, y_{1,n}, y_{2,1}, \ldots, y_{m,n}]^T$ where y_{ij} is observation j for subject i, then the covariance matrix for $\vec{y}$ is expressed as:

$$var(\vec{y}) = \begin{bmatrix} B_1 & 0_{nxn} & 0_{nxn} & \cdots & 0_{nxn} \\ 0_{nxn} & B_2 & 0_{nxn} & \cdots & 0_{nxn} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0_{nxn} & 0_{nxn} & \cdots & \cdots & B_m \end{bmatrix}_{(m*n)x(m*n)}$$

$$\text{where } B_i = \begin{bmatrix} \sigma_{i1}^2 & \rho_{12}\sigma_{i1}\sigma_{i2} & \cdots & \rho_{1n}\sigma_{i1}\sigma_{in} \\ \rho_{12}\sigma_{i2}\sigma_{i1} & \sigma_{i2}^2 & \cdots & \rho_{2n}\sigma_{i2}\sigma_{in} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1n}\sigma_{in}\sigma_{i1} & \rho_{2n}\sigma_{in}\sigma_{i2} & \cdots & \sigma_{in}^2 \end{bmatrix}_{nxn} \quad \text{for subject i}$$

When modeling longitudinal data, the main focus is how the correlation between observations is handled. When correlation is ignored in modeling longitudinal data, the coefficient estimates are incorrect. There are three main approaches to handling such correlations: marginal models, mixed-effect models, and transition models [10]. Similarly to cross-sectional studies, marginal models focus on modeling the marginal mean. In marginal models the correlation is modeled explicitly, where typically it assumed that correlation with-in all units have the same structure, such as AR(1). Mixed-effect models model correlation through the use of individual random effects, such as random intercepts and potentially random slopes, and inferences can still be made on a population level through the use of fixed effects. The nuances of mixed-effect models will be discussed in section 2.1. The final approach is the transition model, which models the correlation structure through a conditional expectation of the outcome based on previous outcomes. It is important to note that all 3 approaches still use covariates in modeling.

1.3 Survey Sampling

Due to cost restrictions and feasibility, many surveys can only sample a proportion of the target population. Through survey sampling methodologies, it can be ensured that the results of surveys are unbiased, so that the insights extracted from them are meaningful. When a survey is performed well, the results of the sample may be considered more accurate than those of the entire population because measurement error is less likely to occur.

When analyzing the population for a survey, 3 different subsets of the population can be considered: the target population, the frame population, and the sampled population. The target population is the set of all units that are part of the population of interest for a study. The frame population is a subset of the target population that is all units in the target population available to sample from. When the target and frame populations are not perfectly aligned, there are units that are not covered in the sampling framed, therefore, it is an incomplete sampling frame. The Sampled population is a subset of the frame population, which are units that have a non-zero probability of being selected. When the sampled population and the frame population are not perfectly aligned, there is a set of units that would not like to participate in the survey. One common mitigation to non-response is through reward incentives for completing a survey, helping to align the frame population and sampled population [9] .

Stratification is a technique for a more efficient and less biased sampling design. It involves dividing the population into non-overlapping strata, where each unit is in exactly one stratum. Typically, strata are divided using a demographic, such as region or age group, to ensure that all units fall into a single stratum. Given population U, the said population can be divided into H strata as follows:

$$U = U_1 \cup U_2 \cup \ldots \cup U_H$$

Clustering is the grouping of units that may be associated with each other. Similarly to stratification, units are only in one cluster. When sampling using clusters, entire clusters are chosen to participate in the survey, for convenience purposes. Sometimes, clustering can be done by dividing each strata into multiple clusters. Complex surveys usually divide the population into stratums (states) and then cluster within each stratum (cities) and randomly select a certain number of clusters per town.[9].

Survey weights are an effective methodology that ensures that inferences from a model are relevant to the target population. For a unit, survey weights are determined by how their demographics are represented in the target population. Units in demographics that are undersampled from the target population will have higher weights, meaning observations from these units will have a higher influence when using survey weights. Units in demographics that are oversampled from the target population will have lower weights, lessening their influence on the model. The process of weighting units can vary from survey to survey in section 3.3, the weighting of units in the Health and Retirement study will be elaborated on.

2 Methodology

2.1 Mixed-Effects Model

2.1.1 Linear Mixed-Effects Model

As mentioned in section 1.2 the mixed effects model contains two fundamental parts: fixed effects and random effects. The fixed effects coefficients are the same for the entire population, capturing population trends. The random effects coefficients are different for each individual, but are drawn from the same distribution. For observation j of individual i the linear mixed-effects model can be written as:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \ldots + \beta_p x_{pij} + \gamma_{0i} + \gamma_1 z_{1ij} + \ldots + \gamma_k z_{qij} + \epsilon_{ij}$$

Or in matrix form:

$$y_i = X_i \beta + Z_i \gamma_i + \epsilon_i$$

Where β is the vector of p fixed effects, X is the design matrix for fixed effects, γ_i is the vector of q random effects, Z is the design matrix for random effects, and ϵ_{ij} is the random error that follows the distribution:

$$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$$

The observation vector y_i comes from a multivariate normal distribution as follows:

$$y \sim MVN(X\beta, V)$$

Where $V = Z\sigma_{\gamma}^{2}X^{T} + \sigma_{\epsilon}^{2}$. The random effects are only the deviation from the population mean; therefore, the mean of the distribution is only dependant on the fixed effects.

Within the random effects of the linear mixed-effects model, there are two parts: random intercepts (γ_{0i}) and random slopes ($\gamma_{1i}, \ldots, \gamma_{qi}$). Random intercepts indicate how far an individual deviates from the sample average at the intercept. This allows different individuals to have a unique intercept, which is important in a study like this because, due to genetic factors and the activity before 50 not being recorded, we can account for variation between individuals being at different starting points.

The random intercept (γ_{0i}) typically follows a normal distribution centered at zero because the fixed random intercept is the average random intercept; thus, the random intercept will be the deviation from the fixed intercept. The resulting intercept for individual i is: $\beta_{0i} + \gamma_{0i}$. In addition, the random intercepts also account for correlation within units observation. The correlation function for two observations of individual i at different times j and k is:

$$Corr(y_{ij}, y_{ik}) = \frac{Cov(y_{ij}, y_{ik})}{\sqrt{Var(y_{ij)}Var(y_{ik})}} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}$$

The resulting correlation between two observations of the same individual is the proportion of total variance explained by the variance within a subject.

Some mixed-effect models only use random intercepts, but others can include random slopes. Random slopes are used for a covariate when its effects vary between individuals. The most commonly used random slope is for time, since time can cause the slopes of individuals to differ at different rates. Other random slopes could be for time-varying covariates if it is expected that each persons effect from said covariate differs. Typically, time-invariant covariates, such as sex or ethnicity, do not require random slopes. Any covariate used for random effects is typically also used for fixed effects. Since the fixed effect slope is the average slope for all observations, the distribution of the random slope is centered at 0, since it is the deviation from the fixed slope for that covariate. The distribution for the vector of random effects, including the slope in individual i is as follows:

$$\gamma_i \sim MVN\left(\vec{0}_{qx1}, \sigma_\gamma = \begin{bmatrix} \sigma^2_{\gamma 0} & \sigma_{\gamma 0}\sigma_{\gamma 1} & \cdots & \cdots \\ \sigma_{\gamma 0}\sigma_{\gamma 1} & \sigma^2_{\gamma 1} & \cdots & \cdots \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \sigma^2_{\gamma q} \end{bmatrix}\right)$$

2.1.2 Weighted Estimation for Mixed Effect Models

Like linear regression, mixed effects models aim to minimize the difference between the actual values and the predicted values. The parameters that need to be estimated are the fixed effect coefficients (β), the variance of the random effects (σ_γ) and the model variance (σ_ϵ). The estimation of σ_γ impacts the estimation of both β and σ_ϵ, so a change in one variable impacts the other 2. Therefore, the parameter estimation must be done simultaneously.

For parameter estimation for mixed effect models, the restricted maximum likelihood estimation method is typically used for unbiased variance estimation through maximizing the restricted likelihood of the joint likelihood after integrating out the fixed effects. However, the inclusion of survey weights induces a pseudo-likelihood that does not correspond to a true probability model, which results in a misspecified likelihood function when using restricted maximum likelihood estimation. This misspecification results in the variance estimation becoming biased, which contradicts the entire purpose of using restricted maximum likelihood estimation [2]

The inclusion of survey weights yields the weighted log-likelihood function as follows:

$$\ell_W(y|x;\beta,\sigma^2_\gamma,\sigma^2_\epsilon) = -\frac{1}{2}\sum_{i=1}^{n} w_i {*} log((nm{*}log(\sigma^2_\epsilon) + m{*}log(|\sigma^2_\gamma|) + \frac{1}{\sigma^2_\epsilon}{*}(y-X\beta)^T\frac{1}{V_i}(y-X\beta)))$$

The analytical solution to this requires taking the determinant and inverse of a symmetric, positive-definite n by n matrix [10]. Therefore, due to this complexity, the Newton-Raphsons algorithm will be used to find the parameter values that maximize this log-likelihood function.

2.1.3 Inference

The weighted likelihood function outlined previously is still misspecified because the estimation is based on a pseudo-likelihood rather than a true likelihood. This results in variance estimates from the Fisher information matrix being inconsistent and does not account for heteroskedasticity from weighting. In addition, in longitudinal data, not all the within-person correlation is captured by the random effects, resulting in unspecified correlation in the model.

The robust standard error allows for the variability in the data to be larger than the specified model through a sandwich estimator. The inclusion of robust standard errors for the inference of fixed effects will allow for further correlations than specified and resolve complications due to the inclusion of survey weights. One result of using robust standard errors is that it allows each individual to have their own variance, which relaxes the assumption of constant variance for the model [1].

2.1.4 Model Evaluation

There are two different approaches to comparing and evaluating mixed-effects models: statistical metrics and model visuals. Statistical metrics are very useful for model comparison due to quantifying the models performance into a value. Visuals can be useful for evaluating how a model fits the data by observing the residuals.

Metrics commonly used for mixed effect models, where the parameters are estimated through maximum likelihood estimation, are AIC and BIC. These two metrics are similar in that they aim to maximize the log likelihood while reducing the number of fixed and random effect parameters present in the model. The difference is that the penalty term in BIC penalizes more heavily for larger datasets, while AIC does not account for larger datasets increasing the log likelihood. For this reason, BIC will be used in this analysis because the dataset used is large.

As discussed previously, the inclusion of survey weights results in the model estimation using the pseudo likelihood. Therefore, the likelihood ratio test statistic does not follow a chi-squared distribution, resulting in the likelihood ratio test not being used in this analysis [4].

Visual metrics are an effective way of visualizing how the model fits the data.

When plotting the residual versus fitted values plot for models, a good fit is when the residuals are scattered around the zero line with no visible pattern. If the points in the residual versus fitted values plot follow a trend, the model is likely not complex enough because of either non-linearity or unexplained variance. As discussed before, because of the use of robust standard errors, heteroskedasticity present in the residuals will not pose a problem for the inferences from the model. In addition, QQ-plots of residuals and random effects can be used, but the inclusion of robust standard errors used in this analysis makes the normality assumptions less strict, allowing us to have heavier tails.

2.2 Longitudinal Survey

2.2.1 Complex Survey Design

Compared to cross-sectional surveys, the survey design for longitudinal surveys is usually more complex and requires more consideration. If an in-person interview is used for a longitudinal survey, it can be very costly and logistically complicated to visit all units in a single wave. One mitigation is to use a complex survey design involving multiple stages of stratification and clustering. When the target population is very large and geographically dispersed, clustering and stratifying the population into regions and randomly sampling those regions is an effective method of reducing costs. A drawback of this is that units from the same cluster could potentially be correlated with each other, as they are more likely to have similar information.

In cross-sectional surveys, if subgroup analysis is of interest, then oversampling from the said subgroups is an effective approach if the selection probabilities are known. For time-invariate subgroups such as race, this method does not pose a problem; however, for time-variate subgroups such as location, this can be an ineffective approach, as people move in and out of the subgroup over time.

2.2.2 Weighting

When performing a longitudinal analysis on all waves, then the latest available weight for a respondent is used for them across all waves. This accounts for sample attrition and non-response bias that accumulates overtime, as well as population changes that occurred over the survey.

Cross-sectional surveys typically have some level of non-response from sampled units, occurring when sampled units decline to be surveyed. This can impose issues for the quality of the survey, but can usually be mitigated through survey weights. Non-response also occurs in longitudinal studies, not just at the first wave, but in all subsequent waves as well. If a respondent drops out, it can be for a multitude of reasons, including but not limited to: difficulty tracking a respondent, respondent fatigue from the interview, or the death of a respondent. Because non-response in later waves is never random, it is critical that as much information as possible about why a respondent dropped-out is provided to make accurate inferences [7].

3 Data Description

3.1 Health and Retirement Study

The Health and Retirement Study (HRS) is a national panel survey conducted from 1992-2022 of US citizens over 50 who do not live in institutions. The HRS is sponsored by the National Institute on Aging (grant number NIA U01AG009740) and is conducted by the University of Michigan [3]. Respondents were surveyed every 2 years, and new cohorts were added to the survey every 6 years. Below is a table identifying cohorts and when they were added to the survey.

Cohort Title	Birth Years	Year Added into Survey
HRS Cohort	1931-1941	1992
AHEAD Cohort	1924 or earlier	1993
Child of Depression	1924-1930	1998
War Baby	1942-1947	1998
Early Baby Boomer	1948-1953	2004
Mid Baby Boomer	1954-1959	2010
Late Baby Boomer	1960-1965	2016
Early Generation X	1966-1971	2022

Table 1: Cohorts of HRS Survey

The following plot is the frequency of each cohort in each wave:

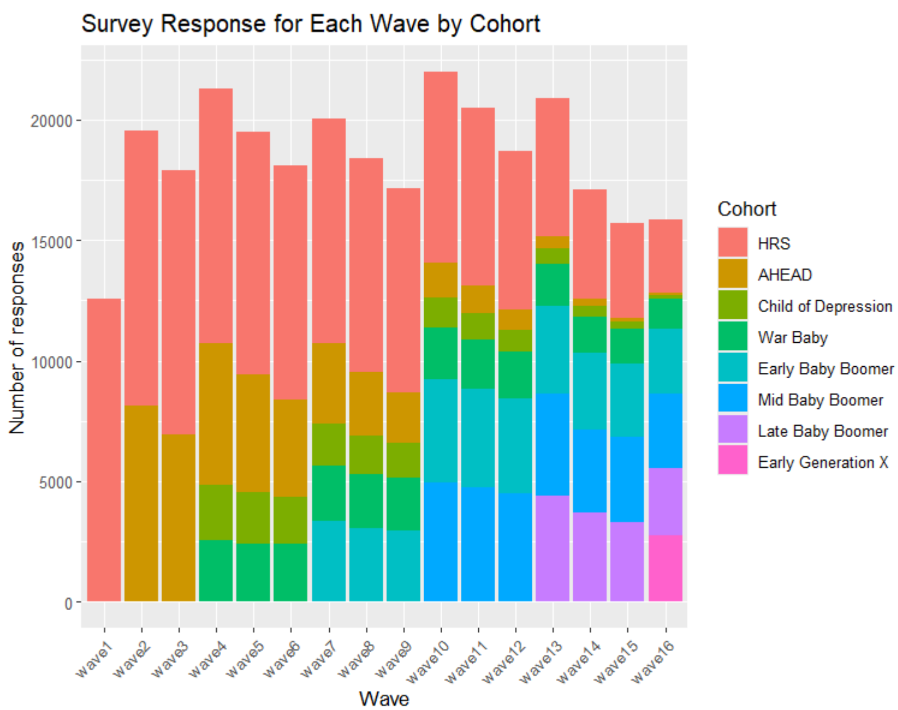

Figure 1: Distribution of Cohorts Across Waves

From the plot, it is evident that there is non-response throughout the survey, as the frequency of people surveyed in a cohort decreases over time. It is recorded if the individuals that did not participate in a wave were alive or passed away. If they did pass away, an exit interview was conducted from a family member or caregiver. The information provided on why a respondent dropped out will be critical to how inferences are made.

When the survey began in 1992, the interviews were conducted only in-person. From 2002, follow-up interviews were conducted by telephone, unless the respondents age is older than 80. Then, starting in 2006, the respondents in follow up interviews were divided into 2 groups, where every wave the groups alternated between face-to-face and telephone interviews. In addition, starting in 2018, respondents had the option to do an online survey instead. Of the eligible respondents who were able to complete the online survey, 40% of them continued to be surveyed using the original methods to compare the results with the respondents who chose to complete the online survey. If a respondent cannot or is unwilling to do an interview themselves, they were given the option to use a proxy for the interview, which was typically a family member.

The sampling design consisted of 4 steps. The first step was to select the primary sampling units (PSU), which are counties in the United States. The counties were divided into metropolitan statistical areas (MSA) and non-MSA counties and then sampled separately from a proportionate probability based on the population of those counties. In the second step, smaller areas in the PSU's, like cities or towns, were sampled. Clusters of homes were selected in the third step, and age-eligible individuals were chosen for interviews in the fourth step.

The HRS surveys included many different questions falling under one of these topics: demographics, health, functional limitations and helpers, financial and housing wealth, income, social security, pensions, health insurance, family structure, retirement plans, employment history, and psycho-social questions.

3.2 Cognition on the 27 Point Scale

The HRS uses several different cognitive tests and questions in their surveys. Due to the HRS being a multi-purpose study and using multiple interviewers due to its large scale, they needed to develop standardized methods to obtain valuable information on cognition in a time efficient manner. At the time the survey began, there was a lack of literature and studies on cognition on a scale as large as the HRS, so the tests needed to be validated by a team of clinicians.

The one used in many literature using the HRS data is the 27 point scale. The score a respondent received is made up of 3 tests, where the maximum score received is 27 points and the minimum is 0. The first test, which is worth 20 points, is immediate and delayed word recall, where the interviewer would read out 10 nouns and the respondent would have to name as many as they can immediately and after a set amount of time. Because there are multiple interviewers and the testing is carried out in the home of the respondents, impacts such as the interviewer reading speed and noise in the testing environment can affect the respondents score. The second test, which is worth 5 points, is counting down from 100 by 7's for 5 iterations. Respondents receive one less point for every error they make, but a mistake made at an earlier iteration will not affect later iterations. For example, if a respondent says 94 on the first iteration, they would lose a point, and then they say 87 on the second iteration, they will not lose a point despite not being on the correct track because 94-7 is 87. This introduces another issue in the methodology: because it is the same test every wave, respondents can practice ahead of time, meaning the test may not be an accurate assessment of cognition. The third test, which is worth 2 points, involves counting backwards by 1's from 20 and receiving a deduction of 1 point for every mistake, with up to 2 points that can be deducted. This test is purposefully designed to be easier than the other tests to identify respondents who are significantly cognitively impaired.

This 27-point cognitive scale is based on the Langa-Weir classification of cognition, defined by researchers at the University of Michigan. Scores between 12 and 27 points are considered normal cognition, 7 to 11 points are considered cognitive impairment without dementia, and 0 to 6 points are considered dementia. These thresholds are not clinical diagnoses but have been validated to meet the criteria for dementia and cognitive impairment in the DSM-III-R and have been reviewed by a group of clinicians [8].

Because the 27-point cognitive scale requires a telephone or in-person interview with a respondent, the tests were not run for web-based and proxy interviews. In addition, the 27-point cognition scale was implemented in the third wave, however, there is other cognition data available prior to the third wave.

3.3 Survey Weights

The survey weights in the HRS are composed of 3 parts: the base weight, non-response adjustment, and post-stratification. The base weight is derived from the sample design, accounting for the inverse of the probability of selection. The HRS survey conducted oversampling of black, Hispanic, and Floridian residents: therefore, their survey weights will be lower as each individual represents a smaller portion of the population. Because of survey attrition, the HRS accounts for the non-response at every wave. This weighting accounts for attrition from death and attrition from voluntarily dropping out of the survey. Post-stratification was performed when the sample was projected to the Current Population Survey (CPS), and weights in the HRS were corrected for the CPS data.

4 Exploratory Data Analysis

4.1 Preprocessing

As stated in section 3.1, in the HRS dataset new cohorts are introduced at various waves in the survey. Because the new cohorts are often younger than the previous cohorts, their cognition at time points is likely to be higher than that of older cohorts at the same time points, even if the cognition score of a member of the new cohort is worse than the cognition score of a member of the old cohort at that age. Therefore, the time point cannot be a fixed effect or random effect in the model unless the analysis is only run on one cohort; however, in doing that we lose lots of valuable information from the previous cohorts. In addition, the age range within a cohort could also distort cognition scores within a cohort. To mitigate these effects, age can be included as a fixed effect and potentially a random effect instead of time, since it puts all individuals in the sample on the same scale. Because the minimum age for the survey is 50, the x axis of the model will be age minus 50, resulting in the intercept being at age 50.

All missing values in the cognition variable were dropped prior to the analysis, due to it being the target variable. The sleeping variables were measured every other time the respondent has been surveyed, therefore the missing values were imputed using the prior value for the sleeping variables. For some of the variables where missing values were likely to be due to a respondent being too ashamed to answer a question such as whether they currently smoke, missing values were imputed with the value a respondent would be more likely to hide. Variables where values were missing likely at random, such as race, were removed from the dataset.

To increase the likelihood that the Newton-Raphson algorithm would converge, the weights were standardized where the sum of the weights would equal the number of individuals. Standardized weights would stabilize the optimization problem, increasing the likelihood of convergence.

4.2 Justification of a Mixed-Effects Model

One of the main purposes of using a mixed-effects model is to effectively model correlated data. If the data is not correlated, then using a regression model estimated through ordinary least squares would be a valid approach to modeling. To model the correlation structure, we can find the empirical correlation for a lag, which is the age difference. To find the empirical correlation at a given lag, all pairs of cognition scores are identified where the age difference is the lag, and the correlation is calculated between all pairs of scores at that lag. Pairs of cognition scores are only used in calculating the correlation if they are from the same individual. Given a vector of scores at age t as x and a vector y, which are scores at age t+u, where x_i and y_i correspond to the same individual, the empirical correlation at lag u is calculated as follows:

$$corr(u) = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}}$$

The correlation between observations in individuals is expected to decrease as the lag increases. To interpret the correlation structure in a correlation plot, a locally estimated scatterplot smoothing (LOESS) line can be fitted to the points in the said correlation plot. LOESS can be used to derive a smooth pattern from data points by fitting a local quadratic regression line for each data point using their neighbours, and then the local quadratic regression lines are put together into a smooth curve. Due to survey attrition and cohorts being added throughout the survey, there are more pairs at smaller lags than at larger lags. To adjust for this discrepancy, a weight based on the number of pairs at a lag can be used to have the LOESS line be closer to points at smaller lags than to larger lags, because there is more confidence in the precision of the correlation at the smaller lags. The correlation plot for age lag using weighted LOESS smoothing is as follows:

Figure 2: Lag Correlation Across Waves

In the correlation plot, there appears to be a slow decay in correlation as the lag increases, indicating that there is strong within-person stability. The correlations at lags 2 through 4 are very high at around 0.7. Individuals cognition scores do not change dramatically on a year-to-year basis, and the measurement is reliable. The correlation is decreasing at a moderate rate, indicating that individuals cognition scores diverge more from the overall trend as aging progresses and that we potentially need random slopes in the mixed-effects models. Because the confidence ribbon of LOESS smoothing line gets significantly wider at around lag 20 large age gaps are not as common and correlation estimates at this lag are more sensitive to noise and less precise. The correlation pattern indicates an auto-regressive(1) correlation structure which can be modeled explicitly in mixed-effects models.

In longitudinal studies, a common plot that is used in exploratory data analysis is the spaghetti plot. A spaghetti plot is where the x axis is time, the y axis is the cognition score, and there is a line for every individual. The distribution at baseline in this plot can indicate if random intercepts will be required, and the requirement for random slopes can be investigated if the rate of cognitive decline is different between individuals.

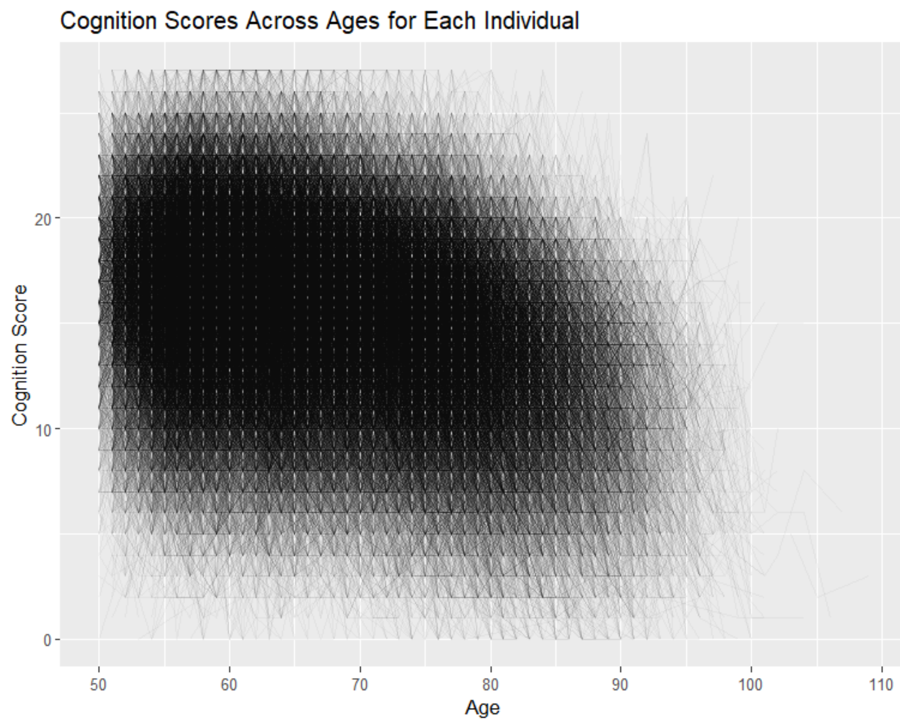

Figure 3: Individual Trajectories of Cognition Across Ages

From the spaghetti plot, it appears that random intercepts are needed for the model, as individuals are at different baselines. The opacity in the spaghetti plot is turned down, so darker regions have more individuals at that point than lighter regions. It appears that at the intercept (50 years old), the individuals are normally distributed, with the points being very dark at a cognition score of 16; then, the further away a score is from 16, the lighter the points become. This would result in a fixed intercept being at a score of 16, and the random intercept is normally distributed at 0 with variance σ_γ^2. The spread of lines changes as the ages increase. First, the spread of the lines gets narrower from age 50 to about age 75, meaning that most people with a higher cognition score at 50 decline at a faster rate than people with a lower cognition score at 50. From age 75 onward, the spread of lines is wider, meaning that individuals are declining at different rates in this age range. This could mean that we need a random slope for age because time could have different effects on different people, because the rate of decline is different for different individuals.

Random effects can also be investigated through distribution plots, such as box plots and violin plots. Box plots identify the quartiles and the range of a continuous variable, and violin plots visualize the distribution across a continuous variable. If the distribution across ages changes it means that random slopes might be required because the cognition is changing at different rates for individuals. Due to ages in the HRS survey ranging from 50 to 109, ages were grouped into bins for easier analysis of the distribution plots. The bins only covered an age range of 5 years, with the exception of all observations of individuals being over 95 years old in one bin, so that important patterns are not missed out. Survey weights are also applied to ensure the insights gained are relevant to the survey weighted mixed effects models.

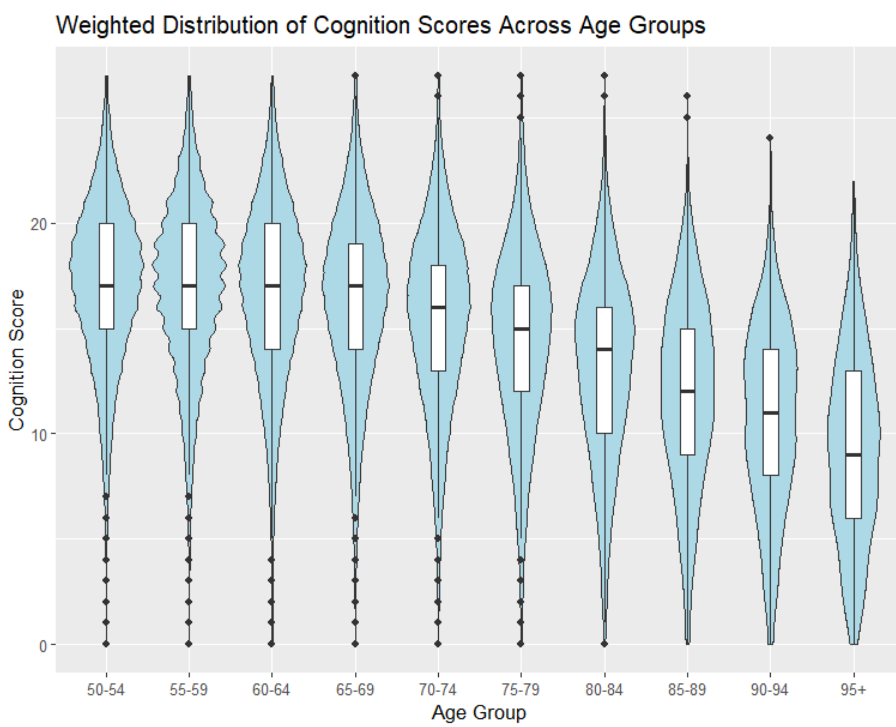

Figure 4: Distribution of Cognition Scores Across Ages

From the violin and box plots, the cognition scores are distributed differently in different age groups. In the three youngest age groups, the scores are less spread out than in the older age groups. This means that random slopes may be needed because changing distributions indicate that individuals experience cognitive decline at different rates due to genetic factors.

4.3 Covariates

Studies have shown that there are non-modifiable and modifiable factors that contribute to cognition, such as race, physical inactivity, smoking, high systolic blood pressure, and depression [6]. The relationship between these factors and cognition in the HRS data will be explored through boxplots to see if there are noticeable differences in the medians and spreads of the distribution, indicating the covariate might be important in determining cognition. The distributions of the levels of the variables can be explored across age groups to analyze the possibility of a random effect or an interaction effect. If the distribution of the covariate levels changes over time, then it can be speculated that the covariate has different effects on individuals, resulting in their cognition declining at different rates. Survey weights will be used in analyzing the covariate boxplots as they will be used in the model. The resulting boxplots can be seen in section 1 of the appendix.

In table 2 below, I have included the resulting covariates that will be used in the initial model, as their boxplots show significant differences in the distributions between the levels of the covariates. The covariates labeled with (R) indicate that they will be modeled with a random slope in the initial model, and the covariates labeled with (I) indicate that they will be modeled with an interaction term with age as well.

Covariate	**Type**	**Description**
Ageminus50 (R)	Demographic	The respondents age minus 50
Currently Smoke (I)	Lifestyle	Whether the respondent currently smokes or not
Diabetic (I)	Health	Whether the respondent is diabetic or not
High Blood Pressure (I)	Health	Whether the respondent has ever had high blood pressure
Psych Problems (I)	Health	Whether the respondent has a diagnosed psychiatric problem
Doesn't Drink (I)	Lifestyle	Whether the respondent currently does not drink
Underweight (I)	Health	Whether the respondent BMI is under 18.5
Hispanic	Demographic	Whether the respondent is Hispanic
Depressed	Health	Whether the respondent scored above a 2 on the CESD test
Education Level	Demographic	Highest level of education achieved (did not complete high school, high school, associates or bachelors, and masters or doctorate)
Exercise Frequency	Lifestyle	Whether the respondent participates in light exercise, multiple times a week, once a week, or never
Heavy drinker	Lifestyle	Whether the respondent drinks over 4 drinks when they do drink
Marriage ended	Lifestyle	Whether the respondent is a widow or divorced
Trouble Falling Asleep	Lifestyle	Whether the respondent has trouble falling asleep multiple times a week
Waking Up Too Early	Lifestyle	Whether the respondent wakes up too early multiple times a week

5 Analysis

5.1 Modeling

To begin the modeling portion, an initial model was run with all the covariates, random effects, and interaction effects specified in the previous section. Some of the coefficient results of the initial model with the significance tests can be seen below in table 3. The full results of the model can be seen in section 2 of the appendix. From the table, we can see that the inclusion of the interaction effects is making most of the corresponding coefficients insignificant.

Covariate	**Coef.**	**p-value**
Currently Smoke	-0.629	<0.001
Currently Smoke * Age	0.030	<0.001
Diabetic	0.051	0.490
Diabetic * Age	-0.142	<0.001
High Blood Pressure	0.299	0.587
High Blood Pressure * Age	-0.0232	<0.001
Psych Problems	-0.113	0.139
Psych Problems * Age	-0.187	<0.001
Doesn't Drink	0.095	0.054
Doesn't Drink * Age	-0.22	<0.001
Underweight	-0.078	0.693
Underweight * Age	-0.026	0.001
Var(Random Slope for Age)	0.010	<0.001

Table 3: Partial results From Model 1

Several reduced models were ran after to see if the BIC and model performance improved. Model 2 removed the random slope for age and all the interaction effects. Model 3 re-added the interaction effect for smoking, in addition the sleeping variables were removed for being insignificant. Model 4 re-added the random slope for age. The coefficient values and significance for all the models can be seen in section 2 of the appendix. In the table 4 the BIC values were compared between all 4 of the models. In model 2 the BIC was significantly lower, meaning we removed too much information. By re-adding the interaction effect for smoking and the random slope for age, we found that the model performed almost as well as the initial model, but with significantly less parameters.

Model	$\ell_{\mathbf{w}}(\mathbf{y}\vert\theta)$	df	BIC
Model 1	-387044.5	28	774423.6
Model 2	-388835.0	21	777921.6
Model 3	-388810.9	20	777860.7
Model 4	-387227.0	21	774704.0

Table 4: BIC Comparisons Between Models

Through analyzing the residual plots, the residual vs fitted values indicate that the linear specification is the correct assumption of the model, with the residuals being randomly scattered, centered at 0 with no pattern. However, observing the residuals vs age plot, there may be mispecification of the linear term for age, as in the older ages, the residuals are no longer centered at 0. The full residual analysis plots can be found in section 3 of the appendix.

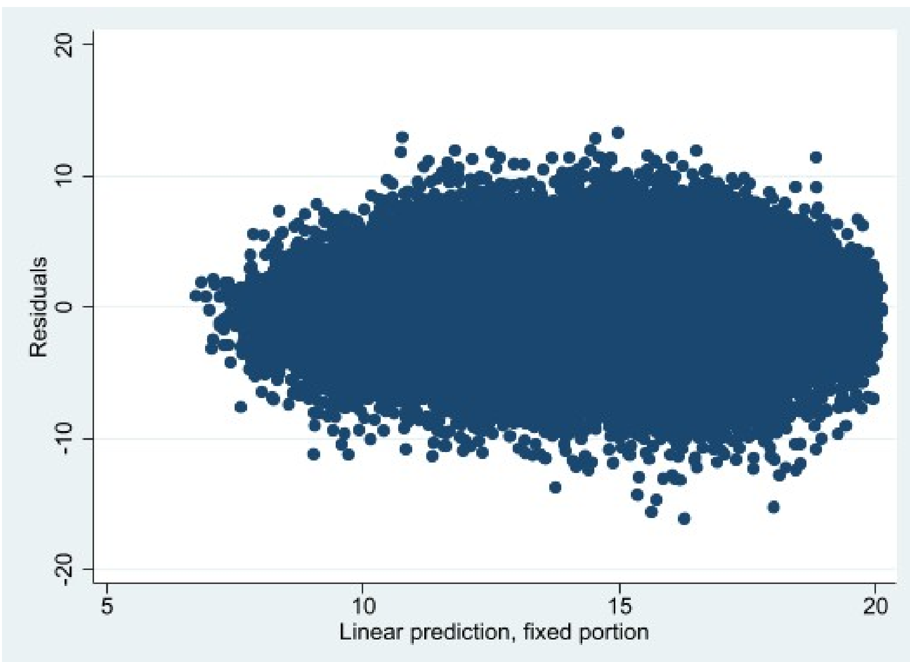

Figure 5: Model 4: Residual vs Fitted Values

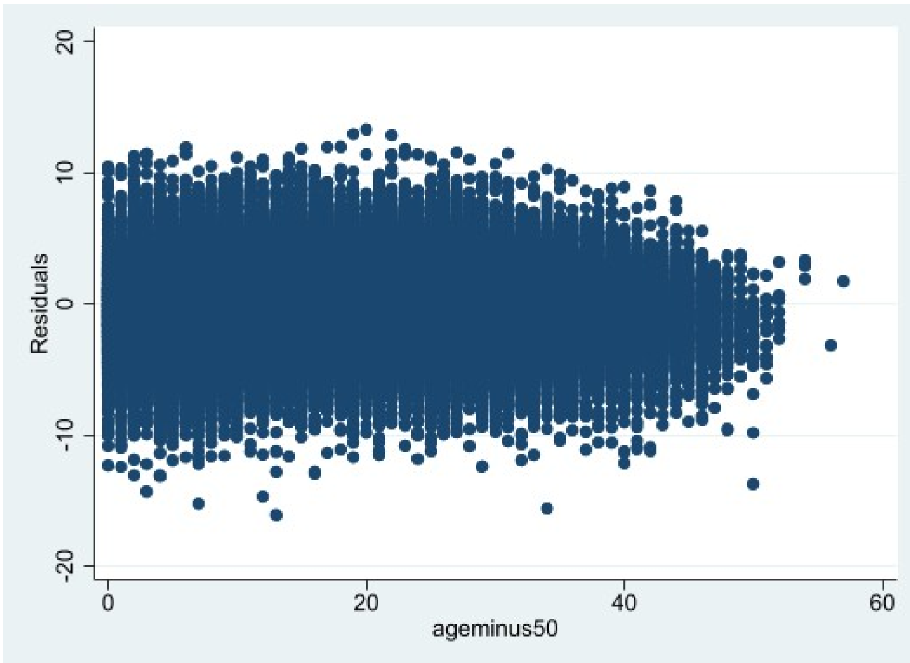

Figure 6: Model 4: Residual vs Age

Because of this potential misspecification a polynomial specification of degree 2 was used in model 5 to assess whether it would be a better fit for the model. The polynomial term was significant, and it improved the fit since the BIC was lowered, as seen in table 5. The polynomial term being significant indicates that cognition declines at a non-linear rate, where it decreases at a faster rate in the later ages.

Model	$\ell_{\mathbf{w}}(\mathbf{y}\vert\theta)$	**df**	**BIC**
Model 4	-387227.0	21	774704.0
Model 5	-385633.2	22	771529.2

Table 5: BIC Comparisons Between Models

The residual plot for model 5 can be seen below. The residuals fan out quite a bit; however, this does not interfere with the inference for the model coefficients because of the use of robust standard errors, allowing individuals to have their own variance. The results and diagnostics from model 5 will be in sections 2 and 3 of the appendix.

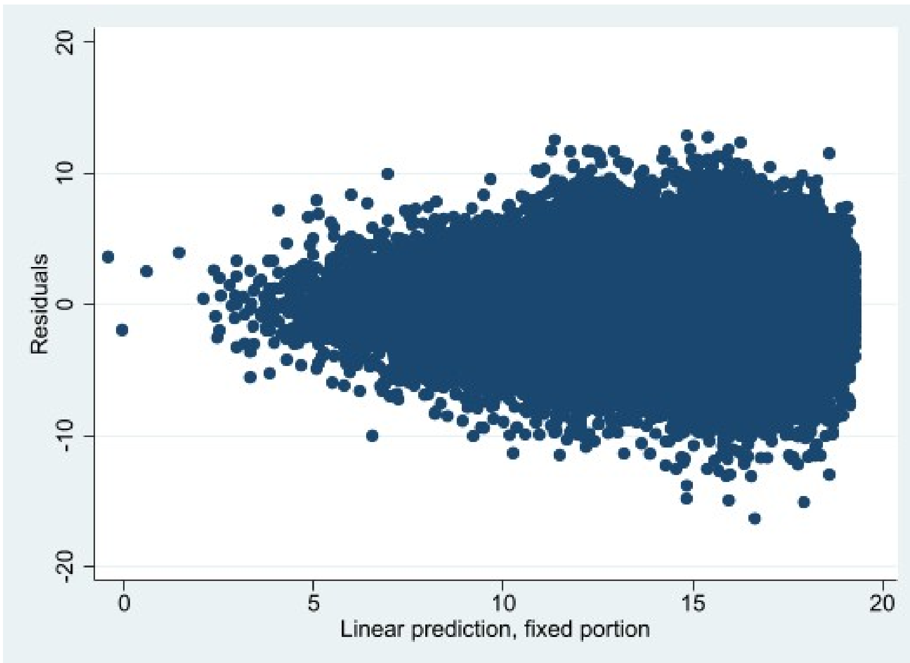

Figure 7: Model 5: Residuals vs Fitted Values

Observing the relationship between age and cognition more closely using a generalized additive model shows that the relationship between cognition and age can be observed in 2 parts. For the population between 50 and 70, the cognition scores seems to decline very slowly, but after age 70, the cognition scores seems to decline at an increasing rate.

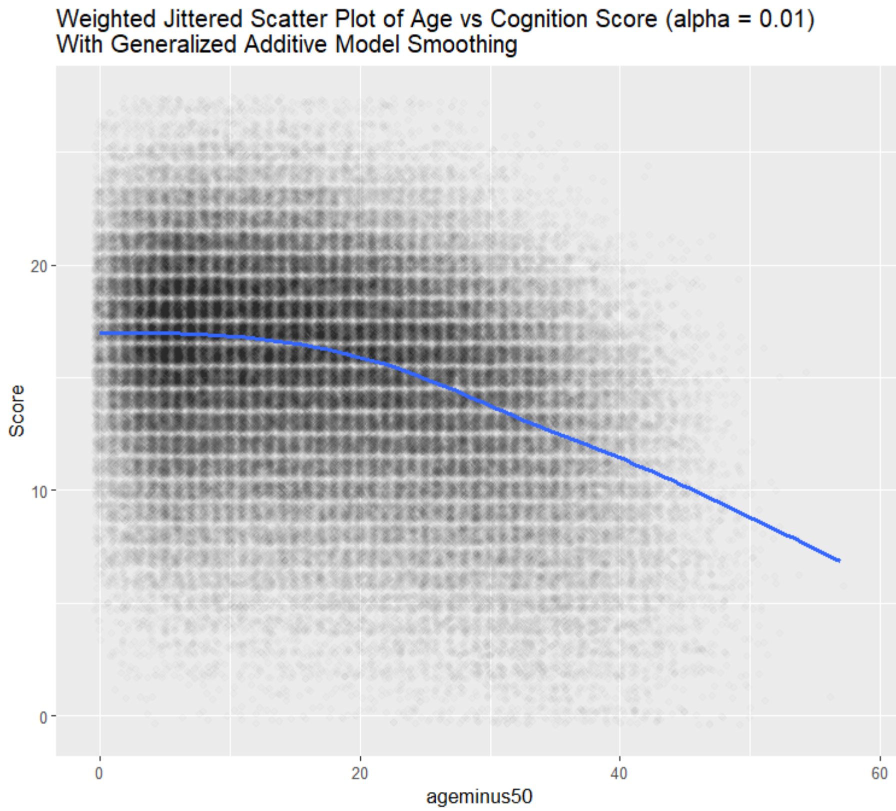

Figure 8: Generalize Additive Model of Age vs Cognition Score

This difference resulted in two additional models being run. Model 6 included only observations below 70, and model 7 included only observations above 70. The BIC is not comparable between these models and other models because the number of observations has changed. The results from these models will be in section 2 and 3 of the appendix.

One way to compare models is through the intra-cluster correlation coefficient (ICC). The ICC shows how much of the total variation of the model can be explained by the random effects. A higher ICC means that there is a higher correlation between observations from the same person and that peoples cognition declines at a consistent rate. From the table below, the ICC between observations below 70 is high, meaning that below the age of 70, peoples cognition declines at a predictable rate, as their deviation from the trend is low.

However, in the above 70 age group, the ICC is much lower, meaning that there is more unpredictability and variance in how cognition declines in older age groups.

	Full Model	**Below 70**	**Above 70**
Var RI	5.6130	5.8100	4.4720
Var RS	0.0097	0.0205	0.0043
Var Resid	6.1932	5.4837	7.2167
ICC	0.4758	0.5153	0.3828

Table 6: Variance of Random Effects in Models

5.2 Results

As mentioned in the introduction, the main inferences of this analysis will be surrounding the lifestyle variables, so that people can adjust modifiable factors that are most linked to cognitive decline. In the figures below I have the 95% confidence intervals for each of the lifestyle coefficient estimates in each of the models.

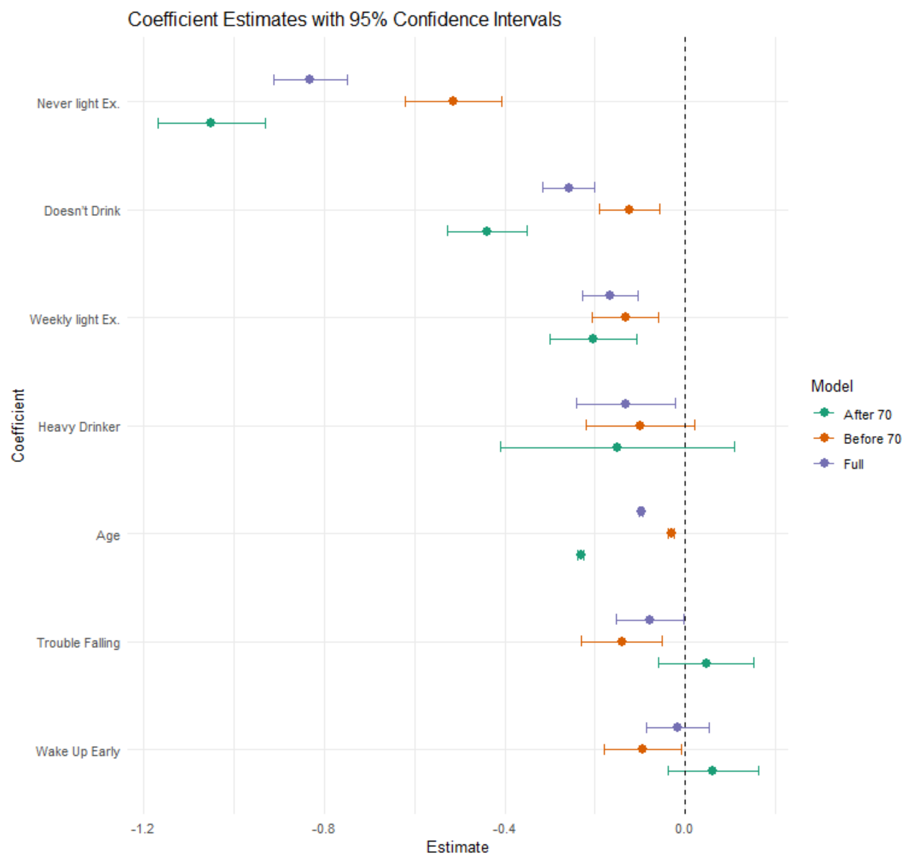

Figure 9: Negative Lifestyle Coefficient Estimates

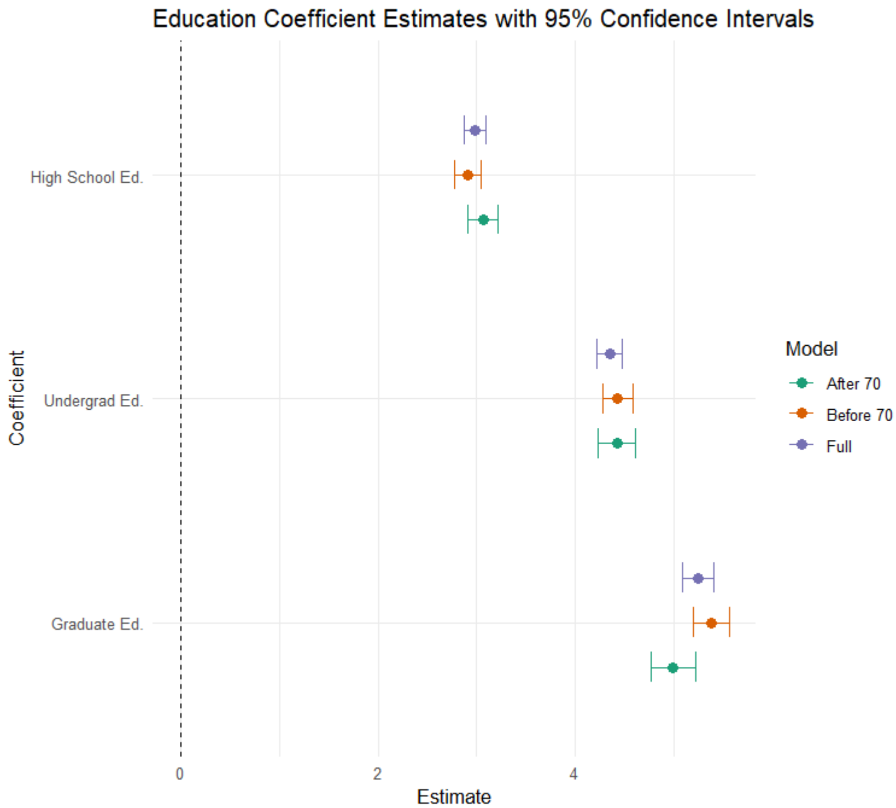

Figure 10: Positive Lifestyle Coefficient Estimates

The most negatively associated lifestyle covariate in all 3 models is engaging in light exercise less than once a week. From this result, we can gather the importance of engaging in light exercise, such as going for a walk, for cognition. From the exploratory data analysis, it can be seen that more intense exercise does not have an impact on cognition and that just getting your body moving, no matter the intensity, will decrease the rate at which your cognition declines.

Another interesting result is the covariate of not drinking being highly negatively associated with cognition, especially for older age groups. This does not necessarily mean that not drinking will worsen cognition, but that there is an association between the two. A reason for this could be that not drinking alcohol can be highly associated with being less social especially in older generations, and social life has a large influence on cognition, In the data there was a social life variable that showed significant impact on cognition, but it was not included in the analysis because it was not measured on many people. However, this supports the fact that a poor social life may be heavily associated with cognitive decline. In addition, the covariate for drinking on average over 4 drinks when a respondent does drink is also associated with cognitive decline, just to a much less significant degree. For cognition, it is ideal to be continue to engage in social activity, especially the older you get, whether it be drinking or not, but if there is drinking to only have a few, as opposed to heavily drinking.

From the 95% confidence intervals, it appears that only sleep issues have a significant negative effect on the younger age groups cognition. This could be related to retirement, as for people who aren't retired, they have to work through the day, resulting in fewer opportunities to catch up on sleep. Whereas people who are retired can catch up on sleep throughout the day if they wake up too early or have issues falling asleep. Therefore, for cognition, it is important when you are employed to ensure you are getting proper sleep.

The only positive lifestyle factor explored is education level, where the higher education levels are more associated with better cognition. This could relate to how stimulating work is, as people with a lower level of education are more likely to work in less stimulating jobs, resulting in a worse cognition overtime. Where as people who have doctorates are more likely to work in more stimulating jobs such as a doctor or professor, where they require higher critical thinking and problem solving through out the day, improving cognition overtime. A takeaway from this is that even if someone has a lower stimulating job, they can still decrease their rate of cognitive decline overtime through stimulating activities like puzzles and reading.

From section 2 of the appendix, it can also be concluded that smoking and high blood pressure are highly associated with lower cognition. So people should be aware of the effects that smoking and nutrition have on decreasing their rate of cognitive decline over time.

6 Conclusion

Through modeling the longitudinal data, it can be determined that the most important factors contributing to cognitive decline are light exercise, social life, and brain stimulating activities.

Future work should aim to develop methodologies for design based inference and model selection criteria for mixed-effects models. Future work in cognitive studies should aim to confirm whether there exists a strong association between nutrition and socially based covariates and cognition, as well as perform a longitudinal analysis on data that includes ages younger than 50, to get a better picture of an individuals life and how it relates to cognition.

7 References

References

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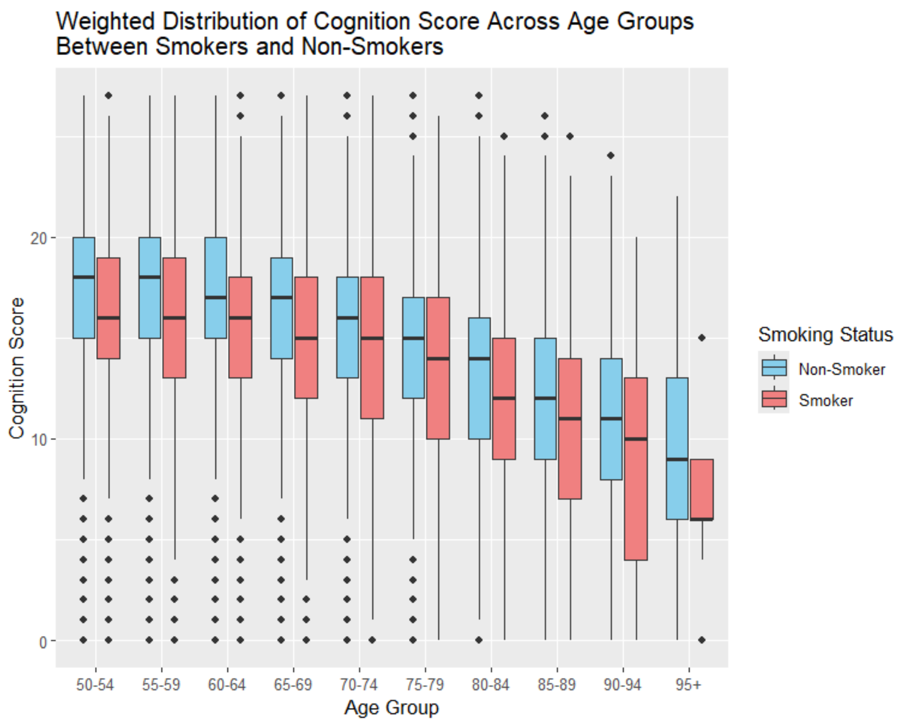
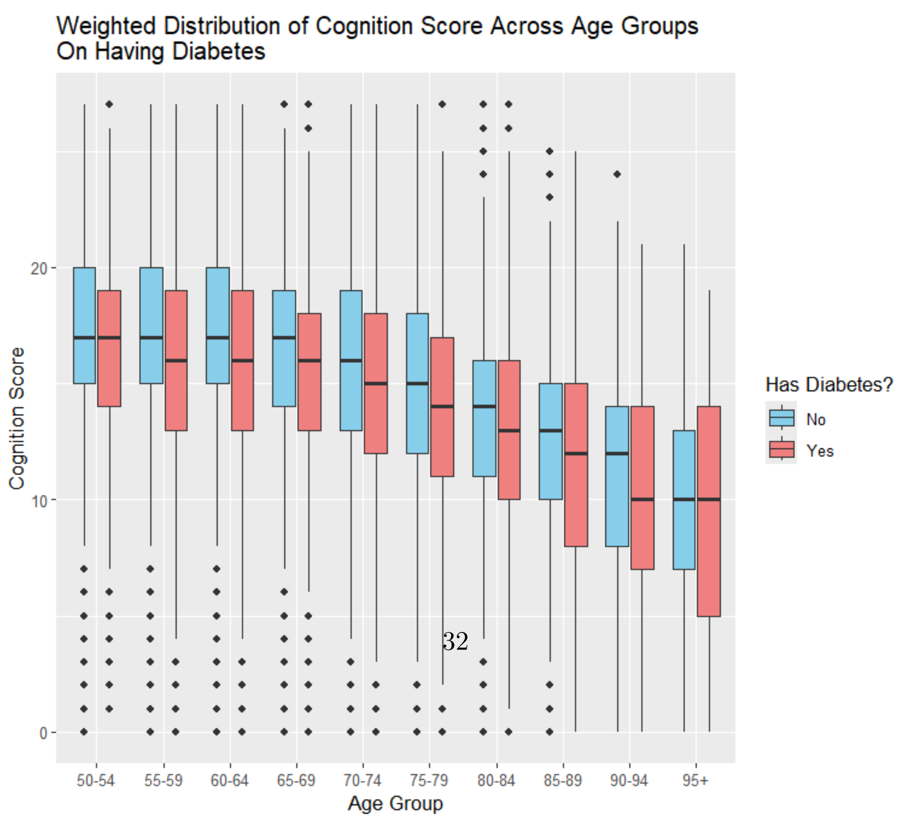
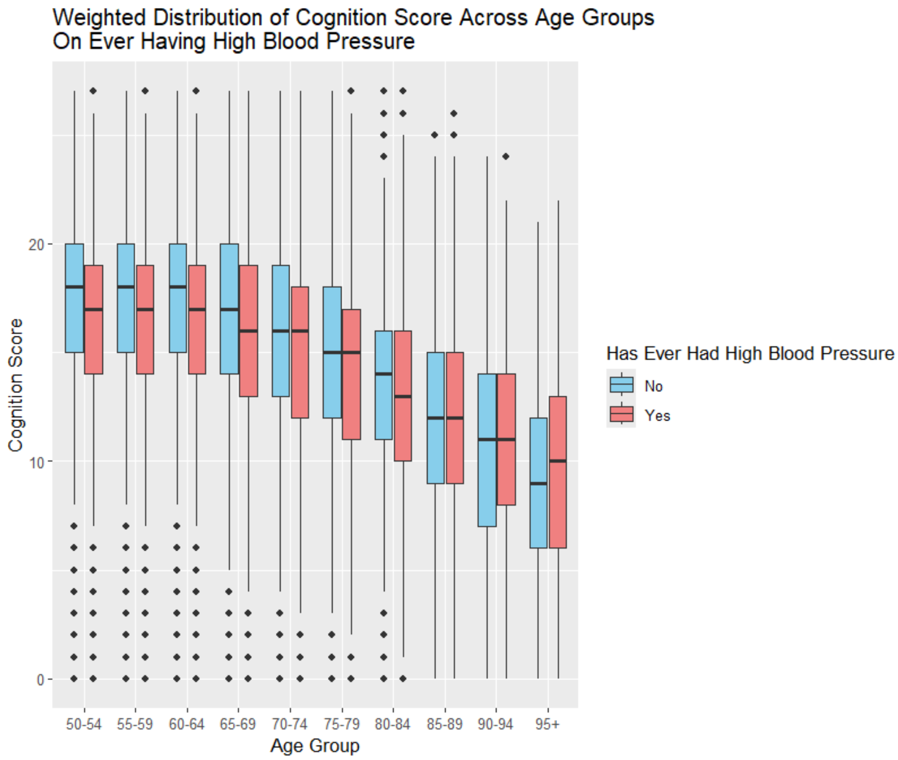
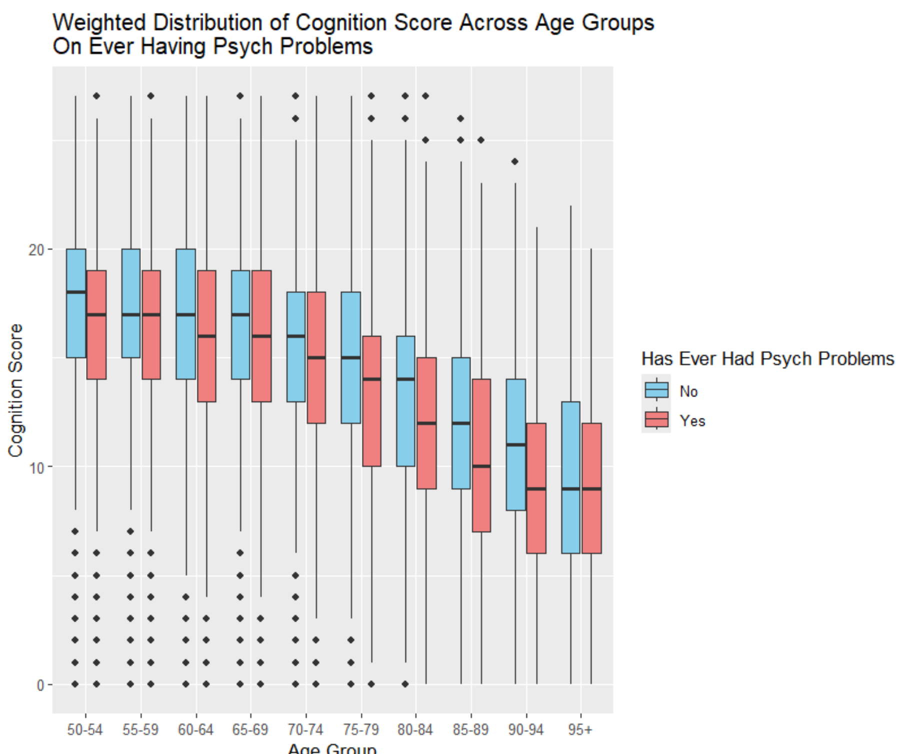
8 Appendix

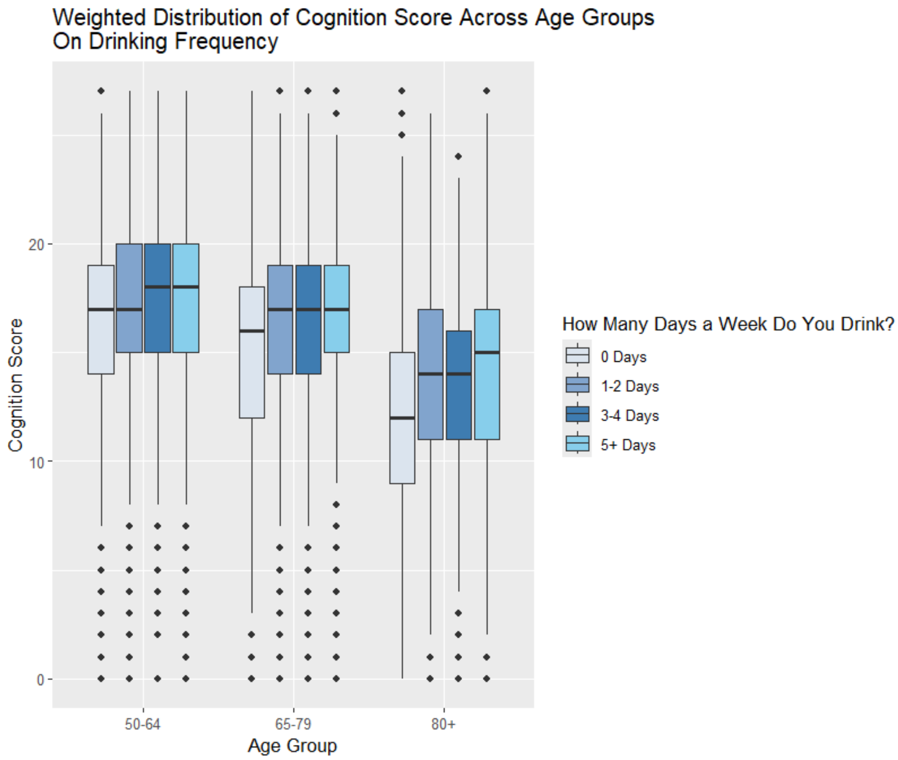
8.1 Section 1: Weighted Covariate Boxplots



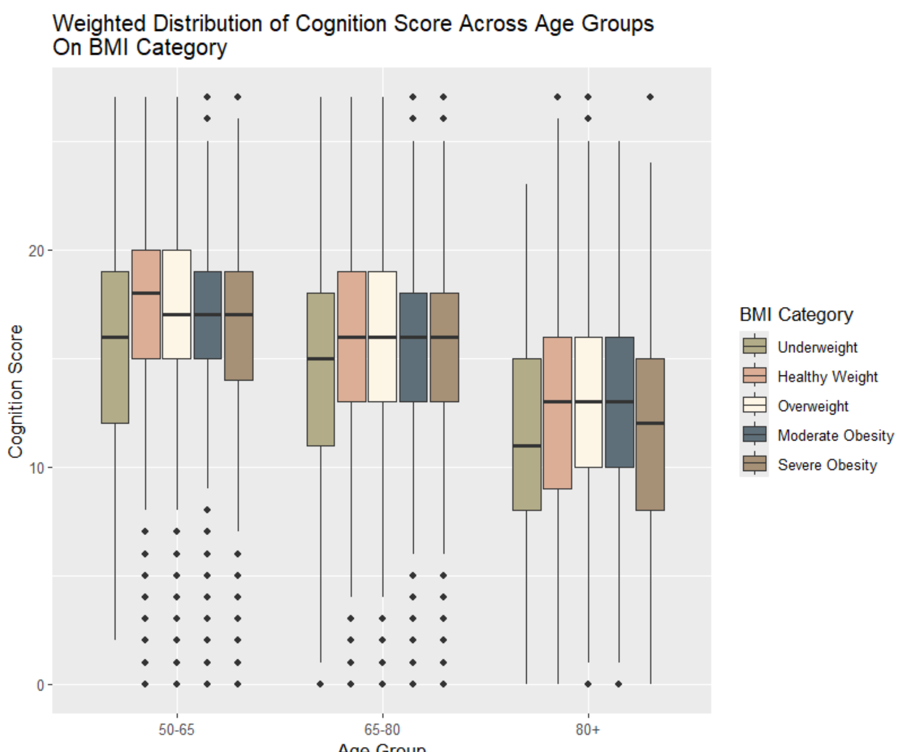



Weighted Distribution of Cognition Score Across Age Groups
On Ever Having High Blood Pressure
Cognition Score
0
10
20
50-54
55-59
60-64
65-69
70-74
75-79
80-84
85-89
90-94
95+
Age Group
Has Ever Had High Blood Pressure
No
Yes

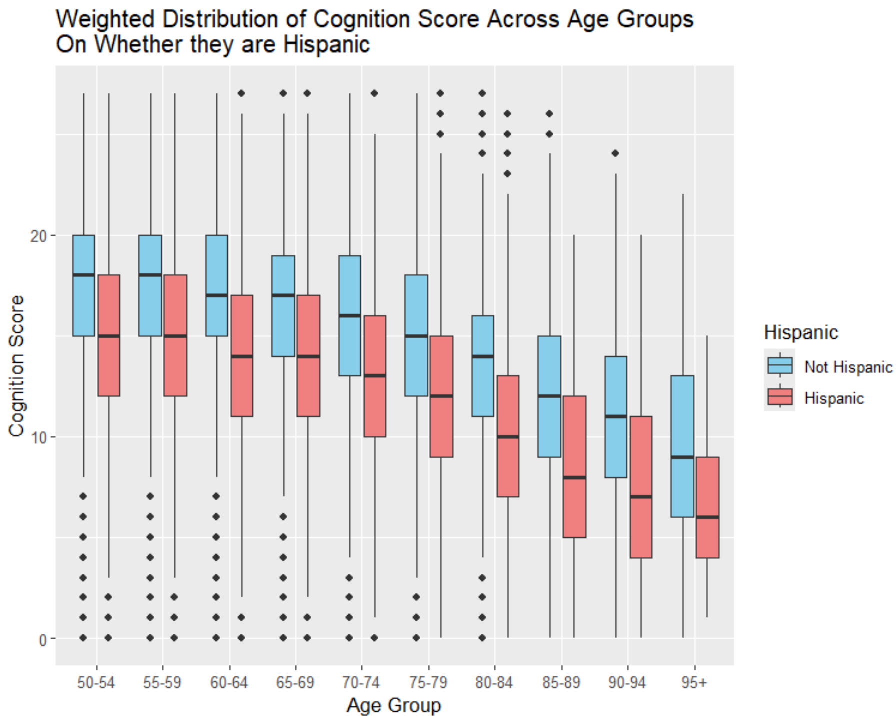

Weighted Distribution of Cognition Score Across Age Groups
On Ever Having Psych Problems
Cognition Score
0
10
20
50-54
55-59
60-64
65-69
70-74
75-79
80-84
85-89
90-94
95+
Age Group
Has Ever Had Psych Problems
No
Yes

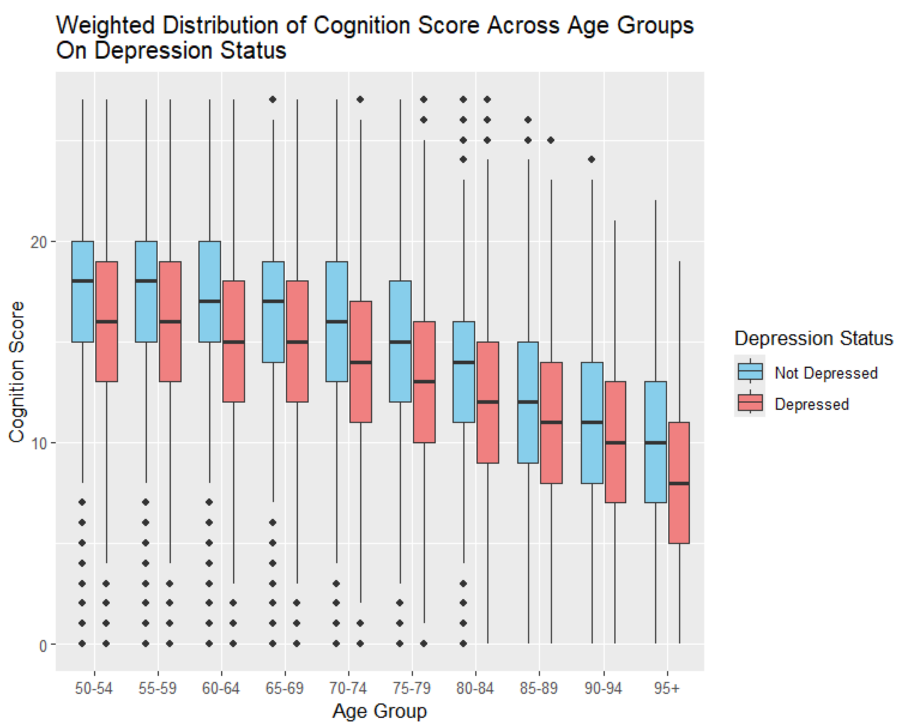


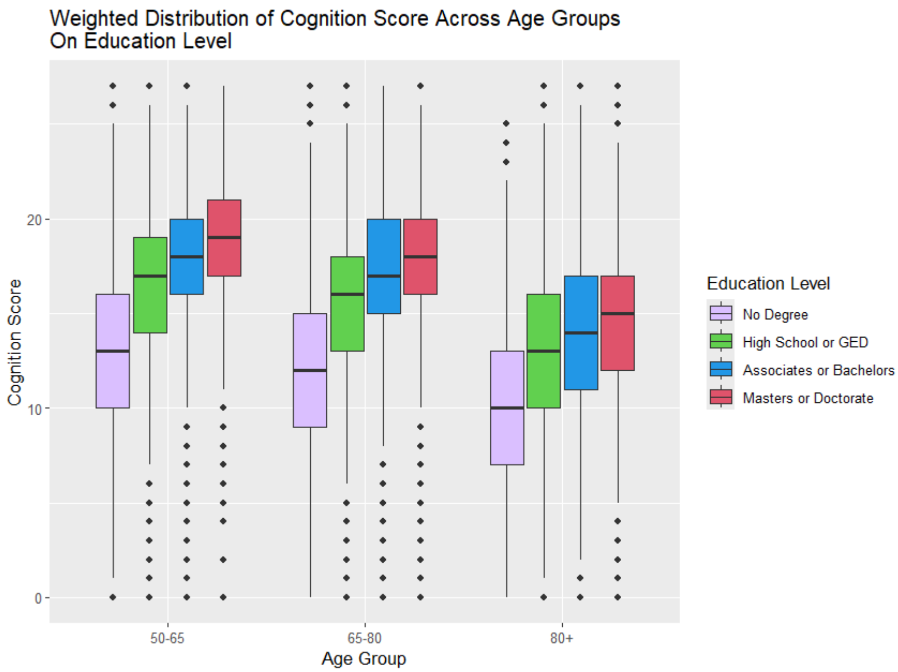
Weighted Distribution of Cognition Score Across Age Groups
On Drinking Frequency
Cognition Score
Age Group
50-64
65-79
80+
0
10
20
How Many Days a Week Do You Drink?
0 Days
1-2 Days
3-4 Days
5+ Days

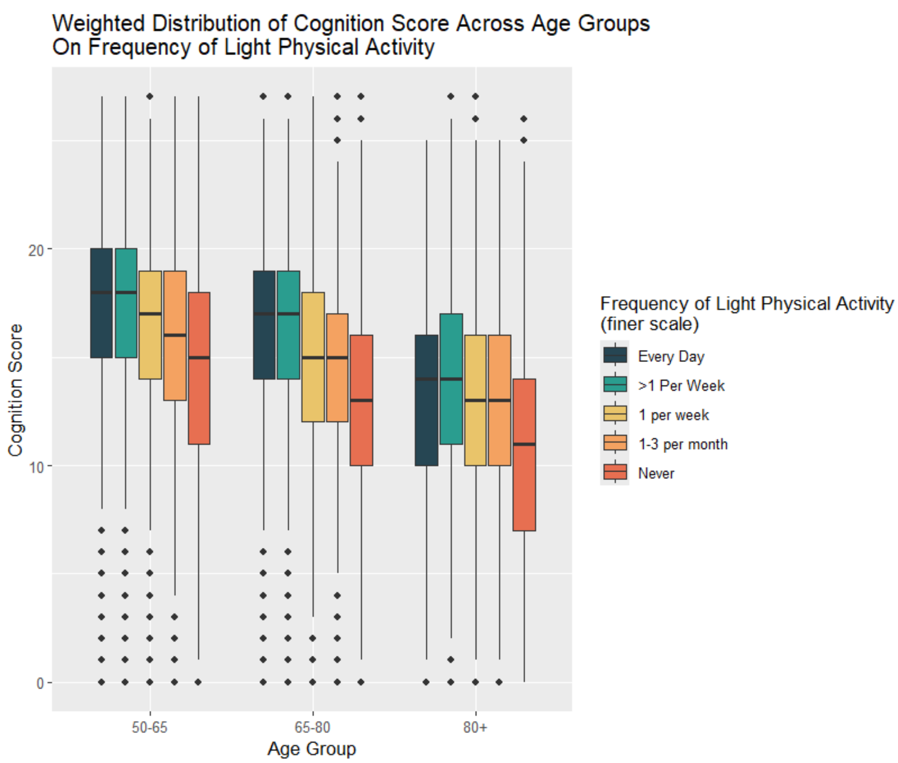


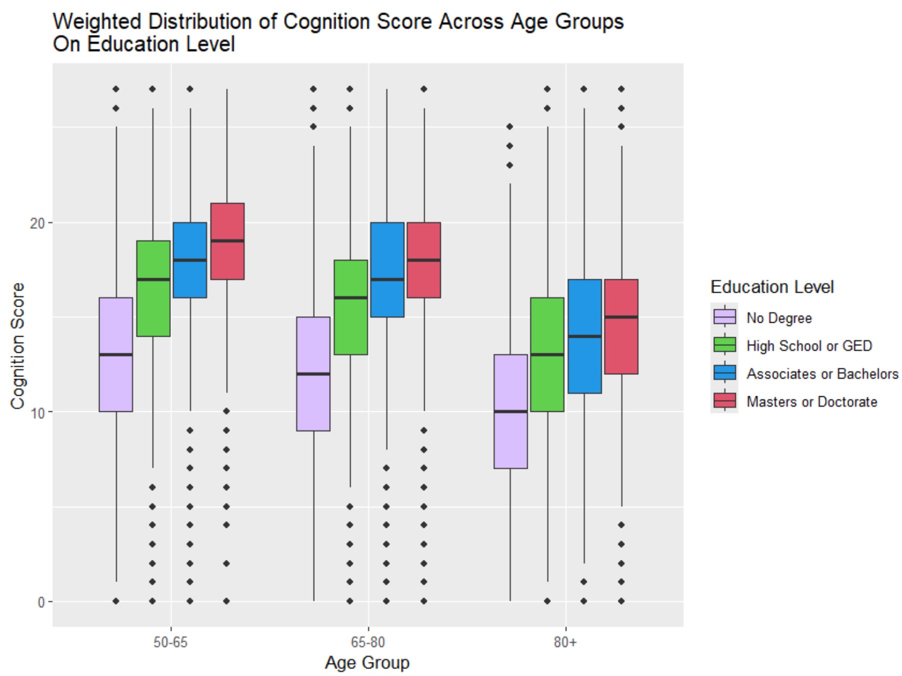
Weighted Distribution of Cognition Score Across Age Groups
On BMI Category
Cognition Score
Age Group
50-65
65-80
80+
0
10
20
BMI Category
Underweight
Healthy Weight
Overweight
Moderate Obesity
Severe Obesity

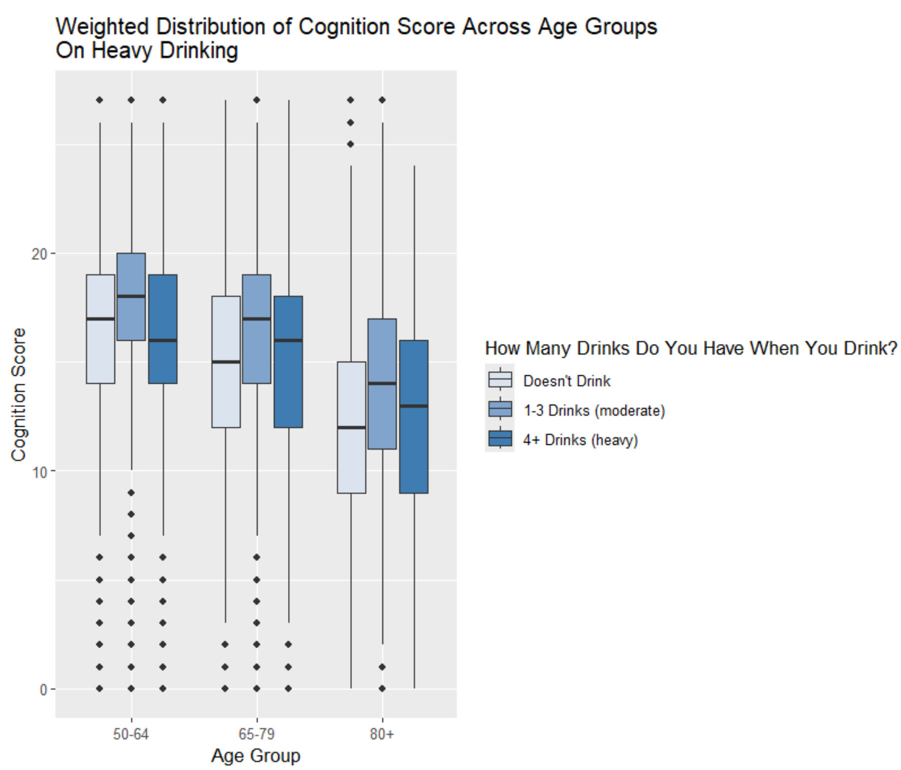

Weighted Distribution of Cognition Score Across Age Groups
On Whether they are Hispanic
Cognition Score
20
10
0
50-54
55-59
60-64
65-69
70-74
75-79
80-84
85-89
90-94
95+
Age Group
Hispanic
Not Hispanic
Hispanic

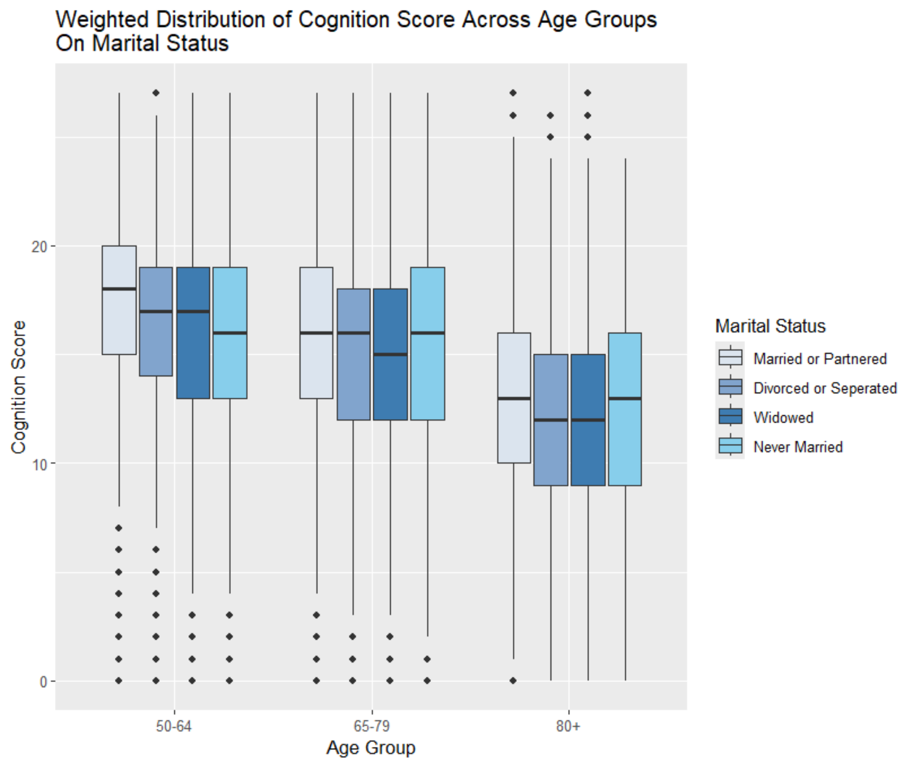

Weighted Distribution of Cognition Score Across Age Groups
On Depression Status
Cognition Score
20
10
0
50-54
55-59
60-64
65-69
70-74
75-79
80-84
85-89
90-94
95+
Age Group
Depression Status
Not Depressed
Depressed

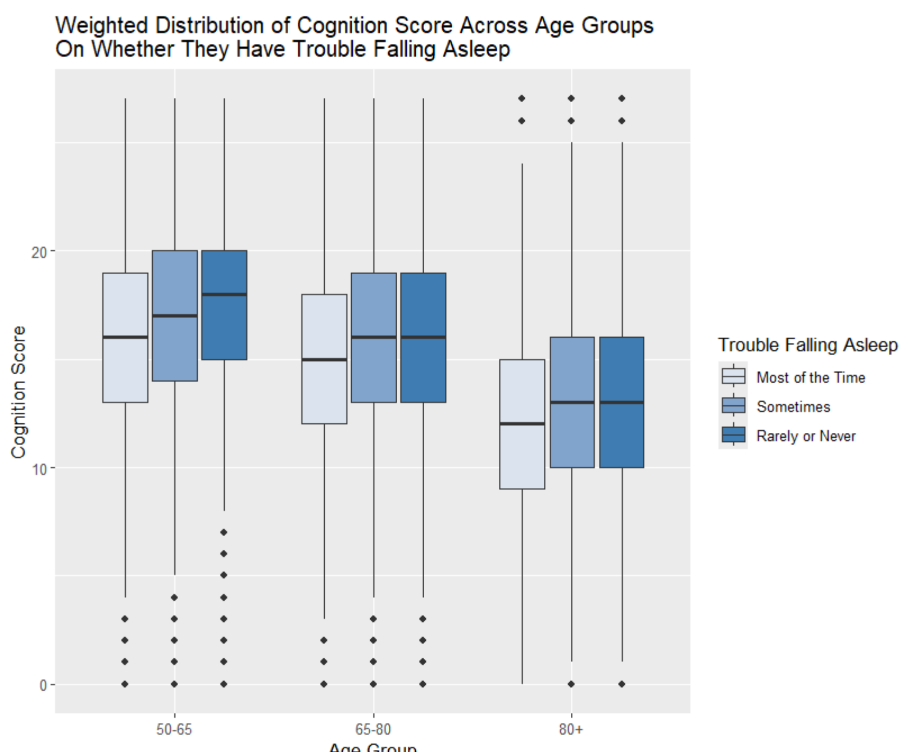

Weighted Distribution of Cognition Score Across Age Groups
On Education Level
Cognition Score
Age Group
50-65
65-80
80+
Education Level
No Degree
High School or GED
Associates or Bachelors
Masters or Doctorate

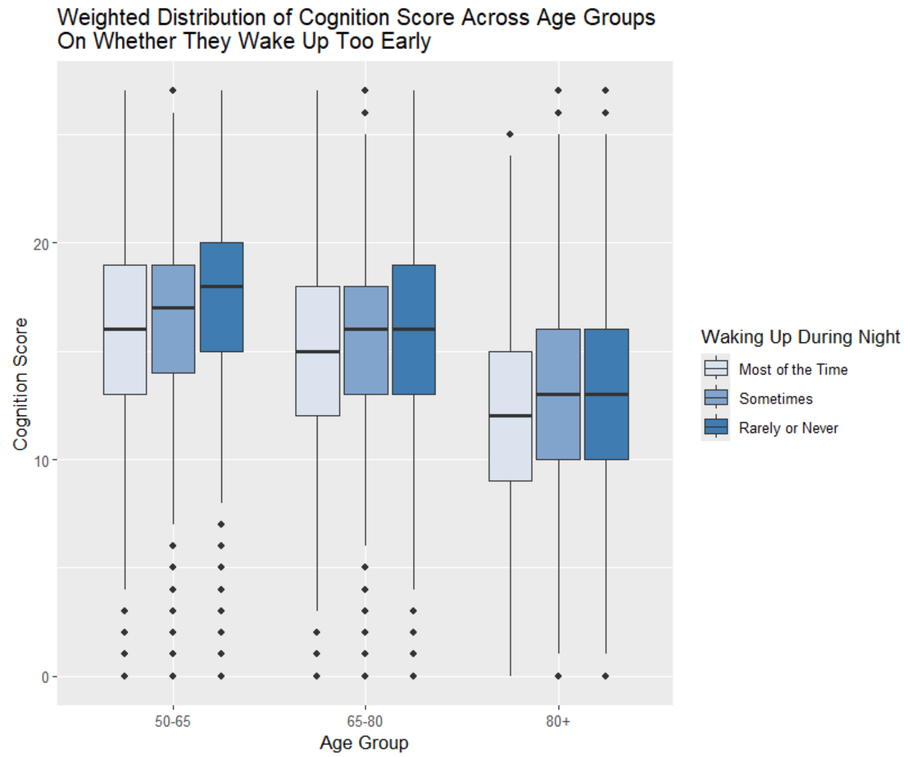

Weighted Distribution of Cognition Score Across Age Groups
On Frequency of Light Physical Activity
Cognition Score
Age Group
50-65
65-80
80+
Frequency of Light Physical Activity
(finer scale)
Every Day
>1 Per Week
1 per week
1-3 per month
Never


Weighted Distribution of Cognition Score Across Age Groups
On Education Level
Cognition Score
0
10
20
50-65
65-80
80+
Age Group
Education Level
No Degree
High School or GED
Associates or Bachelors
Masters or Doctorate

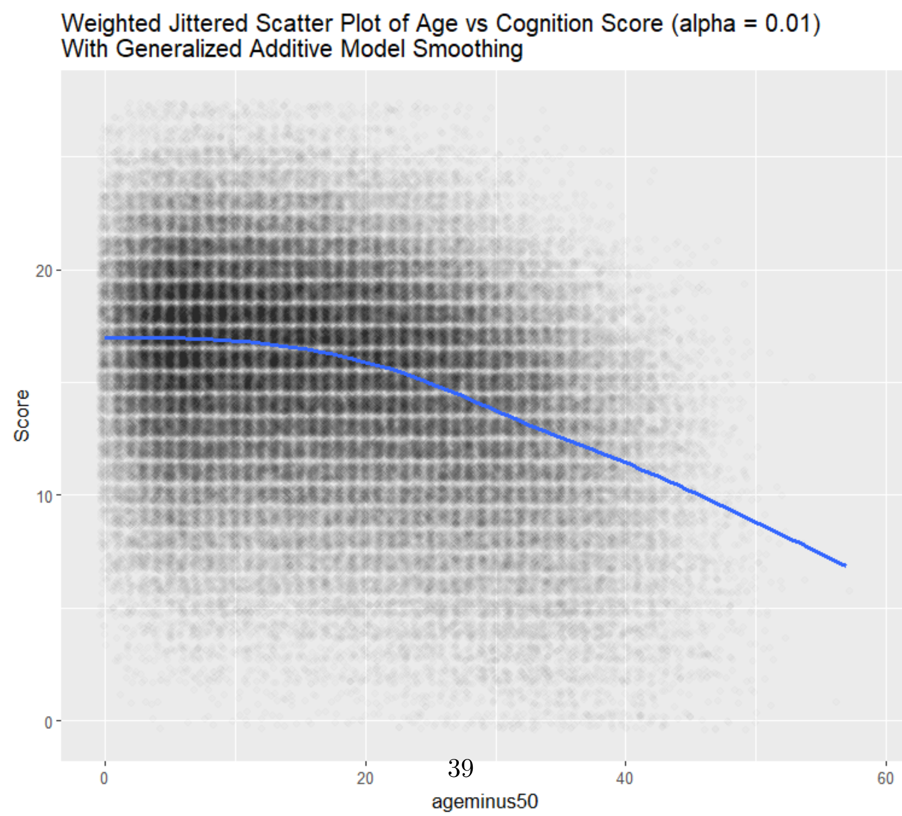

Weighted Distribution of Cognition Score Across Age Groups
On Heavy Drinking
Cognition Score
0
10
20
50-64
65-79
80+
Age Group
How Many Drinks Do You Have When You Drink?
Doesn't Drink
1-3 Drinks (moderate)
4+ Drinks (heavy)


Weighted Distribution of Cognition Score Across Age Groups
On Marital Status
Cognition Score
0
10
20
50-64
65-79
80+
Age Group
Marital Status
Married or Partnered
Divorced or Seperated
Widowed
Never Married


Weighted Distribution of Cognition Score Across Age Groups
On Whether They Have Trouble Falling Asleep
Cognition Score
0
10
20
50-65
65-80
80+
Age Group
Trouble Falling Asleep
Most of the Time
Sometimes
Rarely or Never



Weighted Distribution of Cognition Score Across Age Groups
On Whether They Wake Up Too Early
Cognition Score
20
10
0
50-65
65-80
80+
Age Group
Waking Up During Night
Most of the Time
Sometimes
Rarely or Never



Weighted Jittered Scatter Plot of Age vs Cognition Score (alpha = 0.01)
With Generalized Additive Model Smoothing
Score
20
10
0
0
20
40
60
ageminus50

8.2 Section 2: Model results

Covariate	Coef.	p-value
Age minus 50	-0.067	<0.001
Currently Smoke	-0.629	<0.001
Currently Smoke * Age	0.030	<0.001
Diabetic	0.051	0.490
Diabetic * Age	-0.142	<0.001
High Blood Pressure	0.299	0.587
High Blood Pressure * Age	-0.0232	<0.001
Psych Problems	-0.113	0.139
Psych Problems * Age	-0.187	<0.001
Doesn't Drink	0.095	0.054
Doesn't Drink * Age	-0.22	<0.001
Underweight	-0.078	0.693
Underweight * Age	-0.026	0.001
Hispanic	-0.855	<0.001
Depression	-0.390	<0.001
Highest level of Ed. - Highschool	2.908	<0.001
Highest level of Ed. - Bachelors	4.362	<0.001
Highest level of Ed. - Graduate	5.252	<0.001
Exercise Frequency - <2 a week	-0.162	<0.001
Exercise Frequency - <1 a week	-0.803	<0.001
Heavy Drinker	-0.097	0.083
Marriage Ending	-0.245	<0.001
Trouble Falling Asleep	-0.108	0.003
Wake Up Early	-0.029	0.404
Intercept	14.416	<0.001
Var(Random Intercept)	5.664	<0.001
Var(Random Slope for Age)	0.010	<0.001

Table 7: Results From Model 1

Covariate	**Coef.**	**p-value**
Age minus 50	-0.098	<0.001
Currently Smoke	-0.258	<0.001
Diabetic	-0.122	0.002
High Blood Pressure	-0.182	<0.001
Psych Problems	-0.333	<0.001
Doesn't Drink	-0.259	<0.001
Underweight	-0.696	<0.001
Hispanic	-0.724	<0.001
Depression	-0.413	<0.001
Highest level of Ed. - Highschool	2.984	<0.001
Highest level of Ed. - Bachelors	4.468	<0.001
Highest level of Ed. - Graduate	5.281	<0.001
Exercise Frequency - <2 a week	-0.189	<0.001
Exercise Frequency - <1 a week	-0.932	<0.001
Heavy Drinker	-0.131	0.026
Marriage Ending	-0.278	<0.001
Trouble Falling Asleep	-0.078	0.040
Wake Up Early	-0.018	0.617
Intercept	14.674	<0.001
Var(Random Intercept)	7.323	<0.001

Table 8: Results From Model 2

Covariate	**Coef.**	**p-value**
Age minus 50	-0.101	<0.001
Currently Smoke	-0.581	<0.001
Currently Smoke * Age	0.027	<0.001
Diabetic	-0.124	0.002
High Blood Pressure	-0.186	<0.001
Psych Problems	-0.343	<0.001
Doesn't Drink	-0.261	<0.001
Underweight	-0.716	<0.001
Hispanic	-0.731	<0.001
Depression	-0.417	<0.001
Highest level of Ed. - Highschool	2.989	<0.001
Highest level of Ed. - Bachelors	4.471	<0.001
Highest level of Ed. - Graduate	5.283	<0.001
Exercise Frequency - <2 a week	-0.190	<0.001
Exercise Frequency - <1 a week	-0.933	<0.001
Heavy Drinker	-0.122	0.037
Marriage Ending	-0.282	<0.001
Intercept	14.709	<0.001
Var(Random Intercept)	7.316	<0.001

Table 9: Results From Model 3

Covariate	**Coef.**	**p-value**
Age minus 50 (poly 1)	-488.31	<0.001
Age minus 50 (poly 2)	-269.22	<0.001
Currently Smoke	-0.299	0.021
Currently Smoke * Age (poly 1)	43.775	<0.001
Diabetic	-0.267	<0.001
High Blood Pressure	-0.320	<0.001
Psych Problems	-0.478	<0.001
Doesn't Drink	-0.194	<0.001
Underweight	-0.451	<0.001
Hispanic	-0.753	<0.001
Depression	-0.334	<0.001
Highest level of Ed. - Highschool	3.000	<0.001
Highest level of Ed. - Bachelors	4.512	<0.001
Highest level of Ed. - Graduate	5.285	<0.001
Exercise Frequency - <2 a week	-0.155	<0.001
Exercise Frequency - <1 a week	-0.685	<0.001
Heavy Drinker	-0.076	0.183
Marriage Ending	-0.113	0.001
Intercept	12.917	<0.001
Var(Random Intercept)	6.687	<0.001
Var(Random Slope for Age)	341495.5	<0.001

Table 10: Results From Model 4

Covariate	**Coef.**	**p-value**
Age minus 50	-0.101	<0.001
Currently Smoke	-0.661	<0.001
Currently Smoke * Age	0.033	<0.001
Diabetic	-0.176	<0.001
High Blood Pressure	-0.285	<0.001
Psych Problems	-0.386	<0.001
Doesn't Drink	-0.218	<0.001
Underweight	-0.653	<0.001
Hispanic	-0.854	<0.001
Depression	-0.401	<0.001
Highest level of Ed. - Highschool	2.913	<0.001
Highest level of Ed. - Bachelors	4.351	<0.001
Highest level of Ed. - Graduate	5.245	<0.001
Exercise Frequency - <2 a week	-0.167	<0.001
Exercise Frequency - <1 a week	-0.832	<0.001
Heavy Drinker	-0.132	0.019
Marriage Ending	-0.266	<0.001
Intercept	14.865	<0.001
Var(Random Intercept)	5.613	<0.001
Var(Random Slope for Age)	0.010	<0.001

Table 11: Results From Model 5

Covariate	**Coef.**	**p-value**
Age minus 50	-0.031	<0.001
Currently Smoke	-0.276	<0.001
Diabetic	-0.266	<0.001
High Blood Pressure	-0.368	<0.001
Psych Problems	-0.333	<0.001
Doesn't Drink	-0.124	<0.001
Underweight	-0.460	0.001
Hispanic	-0.833	<0.001
Depression	-0.349	<0.001
Highest level of Ed. - Highschool	2.906	<0.001
Highest level of Ed. - Bachelors	4.429	<0.001
Highest level of Ed. - Graduate	5.377	<0.001
Exercise Frequency - <2 a week	-0.133	<0.001
Exercise Frequency - <1 a week	-0.514	<0.001
Heavy Drinker	-0.100	0.104
Marriage Ending	-0.185	<0.001
Trouble Falling Asleep	-0.140	0.002
Wake Up Early	-0.095	0.030
Intercept	14.146	<0.001
Var(Random Intercept)	0.020	<0.001
Var(Random Slope for Age)	5.810	<0.001

Table 12: Results From Model 6

Covariate	**Coef.**	**p-value**
Age minus 50	-0.232	<0.001
Currently Smoke	-0.267	0.003
Diabetic	-0.359	<0.001
High Blood Pressure	-0.162	0.002
Psych Problems	-0.564	<0.001
Doesn't Drink	-0.440	<0.001
Underweight	-0.496	<0.001
Hispanic	-0.741	<0.001
Depression	-0.390	<0.001
Highest level of Ed. - Highschool	3.066	<0.001
Highest level of Ed. - Bachelors	4.424	<0.001
Highest level of Ed. - Graduate	4.992	<0.001
Exercise Frequency - <2 a week	-0.203	<0.001
Exercise Frequency - <1 a week	-1.050	<0.001
Heavy Drinker	-0.151	0.256
Marriage Ending	-0.021	<0.666
Trouble Falling Asleep	0.046	0.387
Wake Up Early	0.062	0.226
Intercept	18.566	<0.001
Var(Random Intercept)	0.004	<0.001
Var(Random Slope for Age)	4.472	<0.001

Table 13: Results From Model 7

8.3 Section 3: Residual Analysis

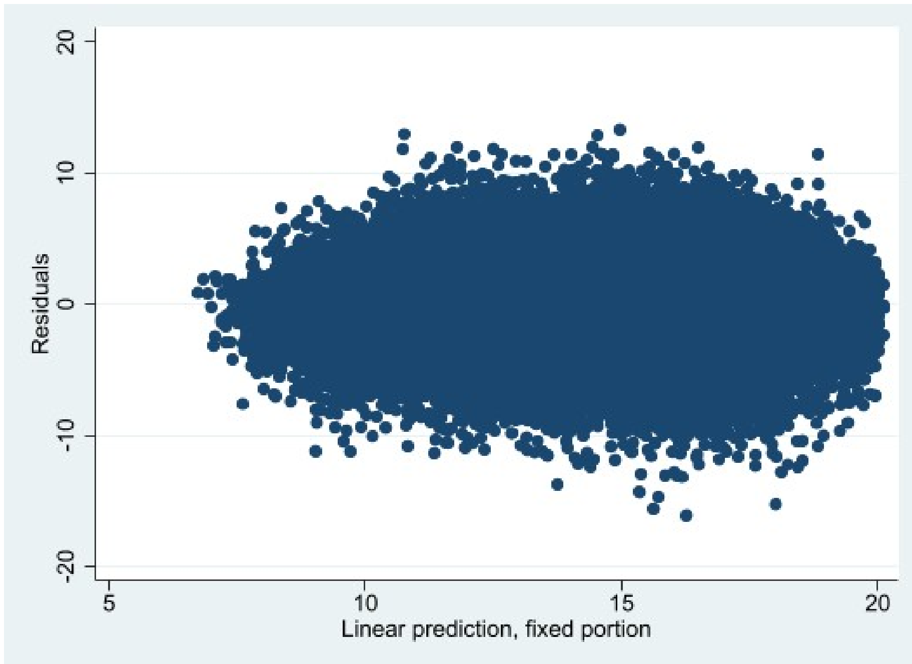


Figure 11: Model 4 Residual vs Fitted

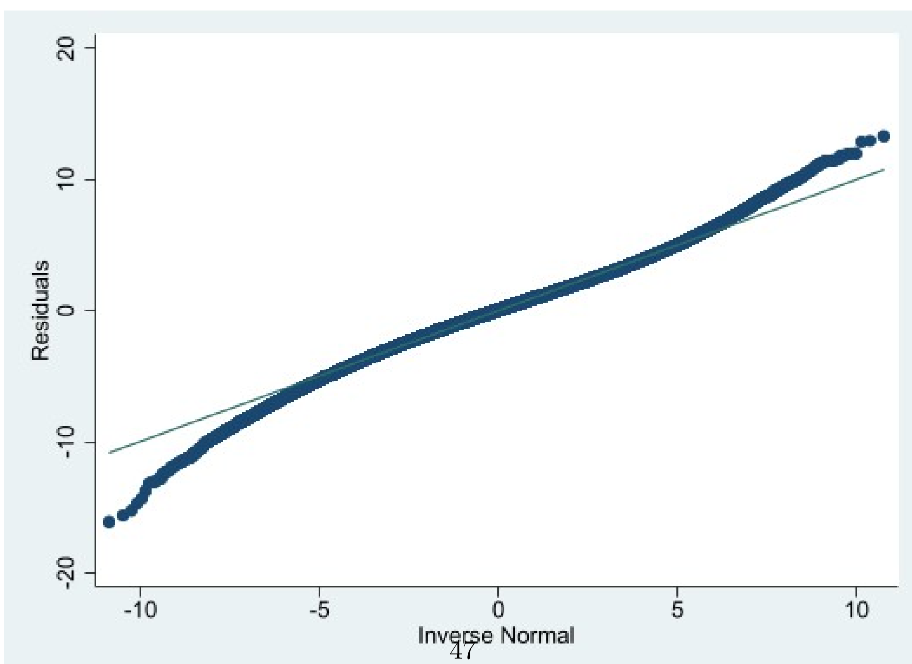


Figure 12: Model 4 Residual Quantile Plot

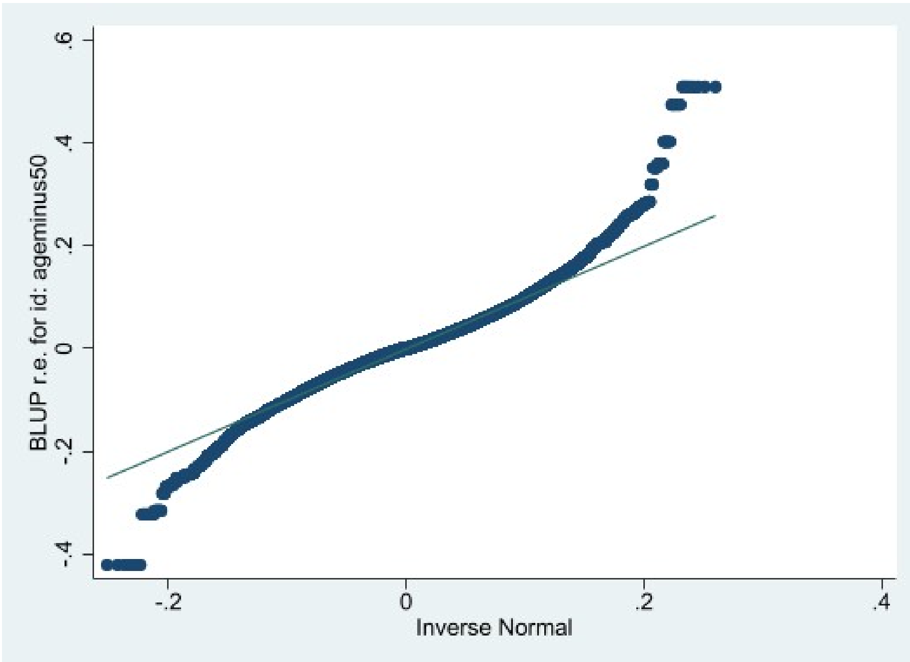


Figure 13: Model 4 Random Intercept Quantile Plot

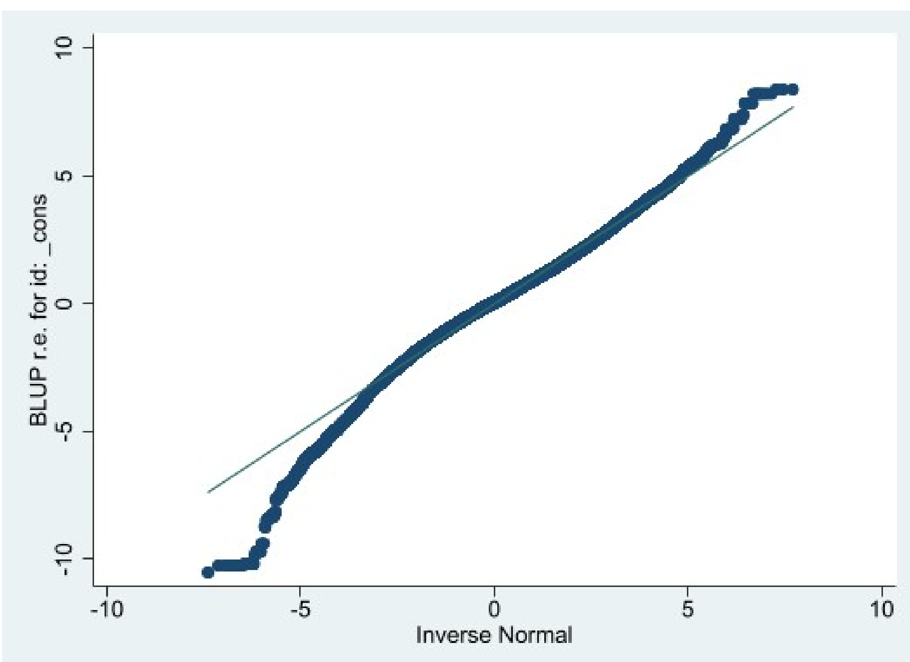


Figure 14: Model 4 Random Slope Quantile Plot

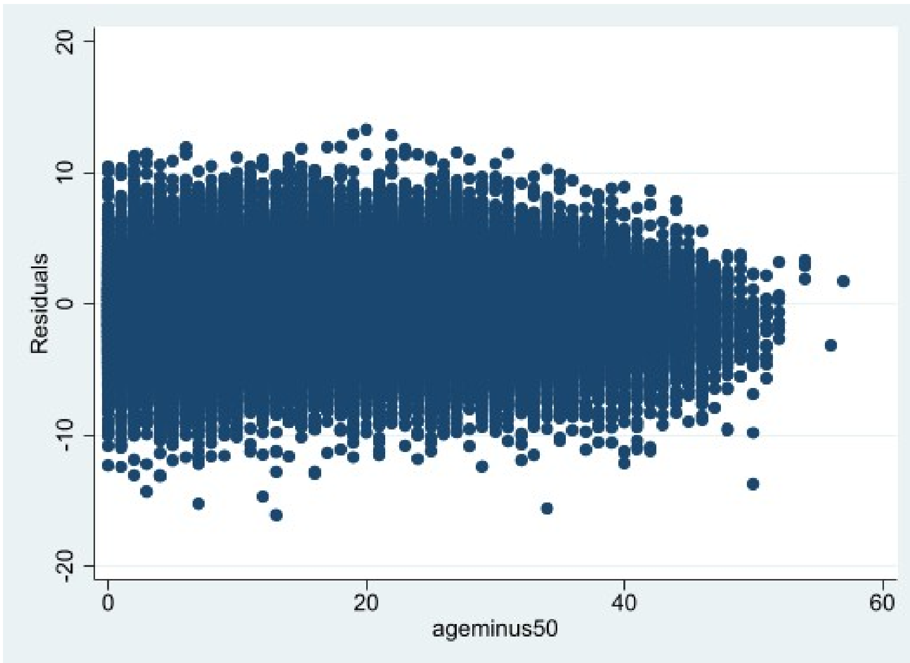


Figure 15: Model 4 Residual vs Age

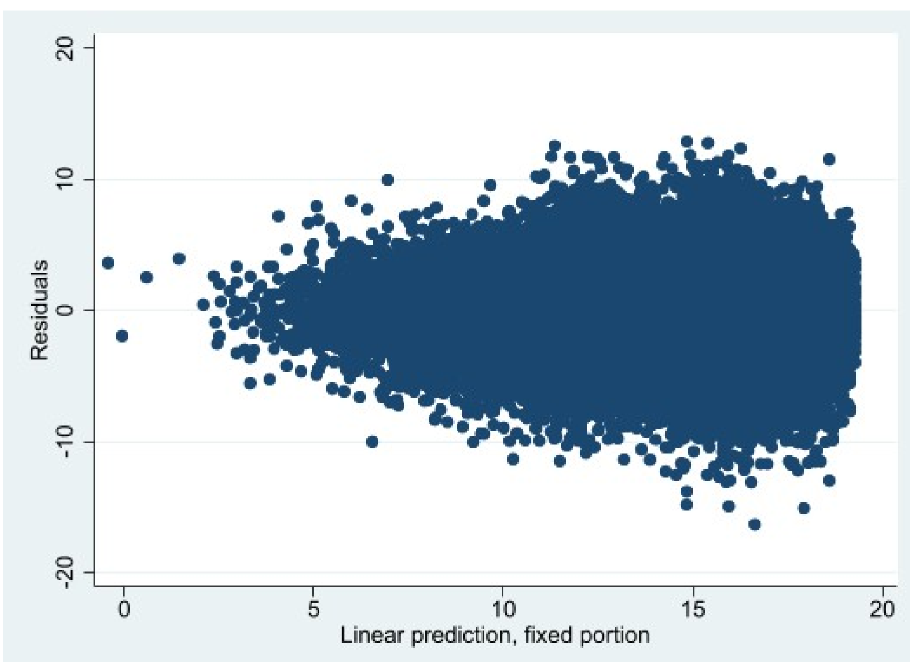


Figure 16: Model 5 Residual vs fitted values

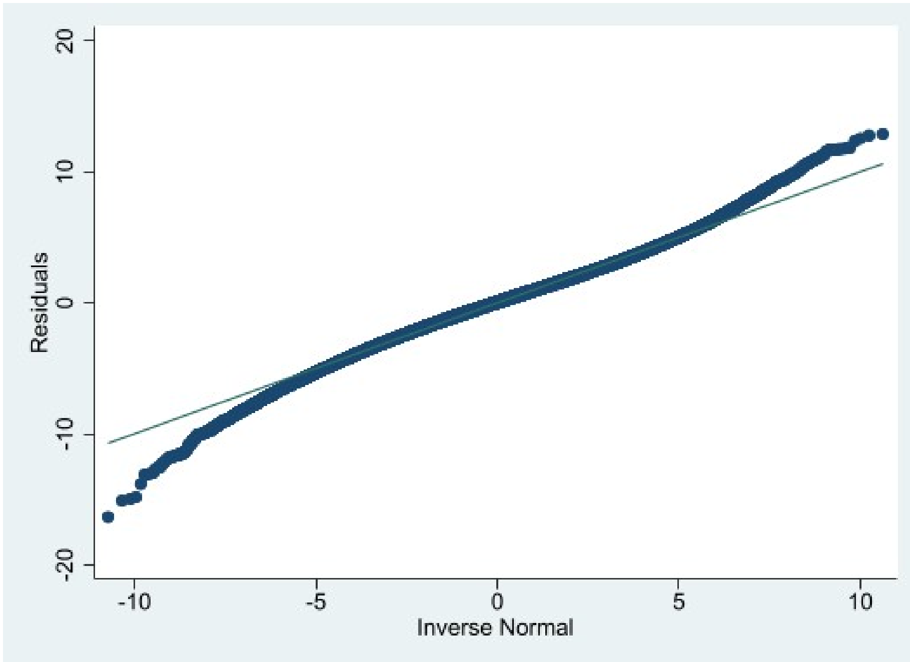


Figure 17: Model 5 Residual Quantile Plot

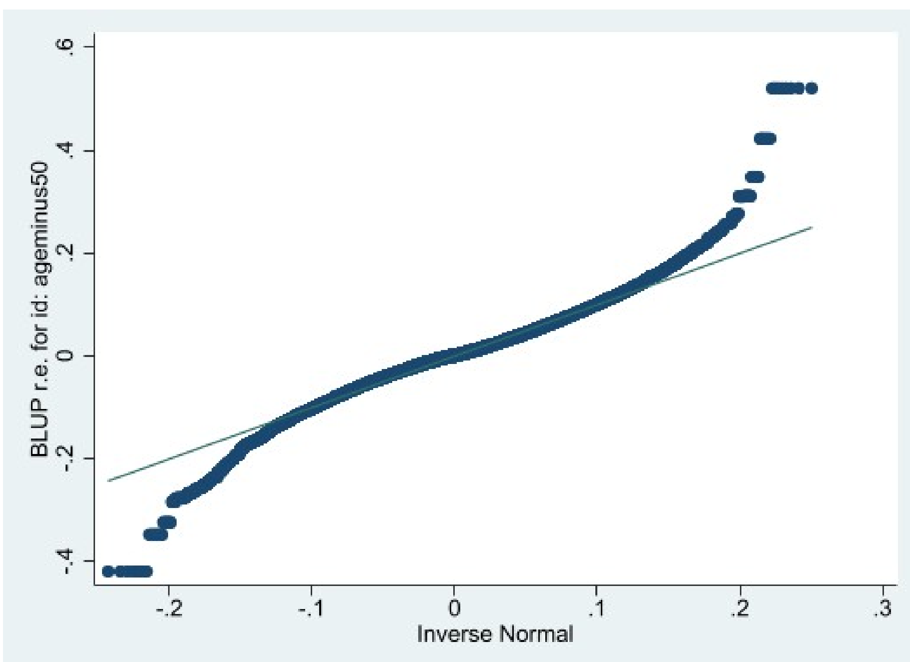


Figure 18: Model 5 Random Intercept Quantile Plot

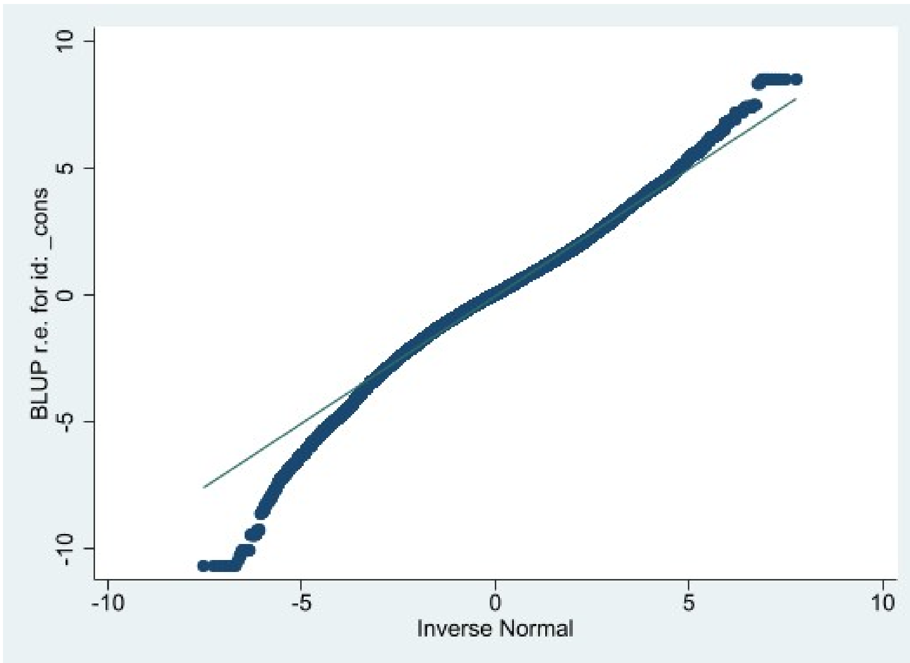


Figure 19: Model 5 Random Slope Quantile Plot

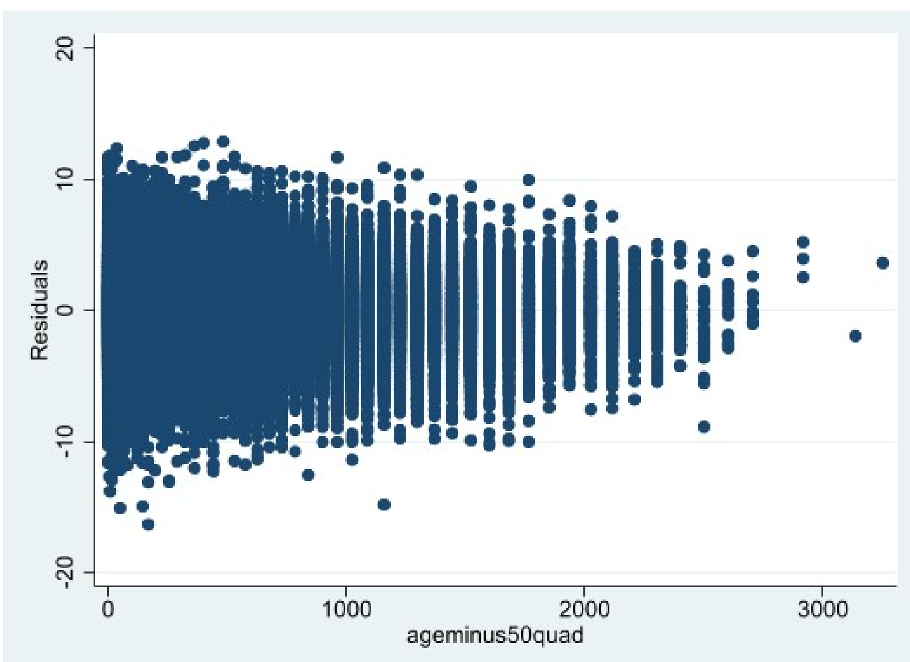


Figure 20: Model 5 Quadratic Age vs Resdiuals

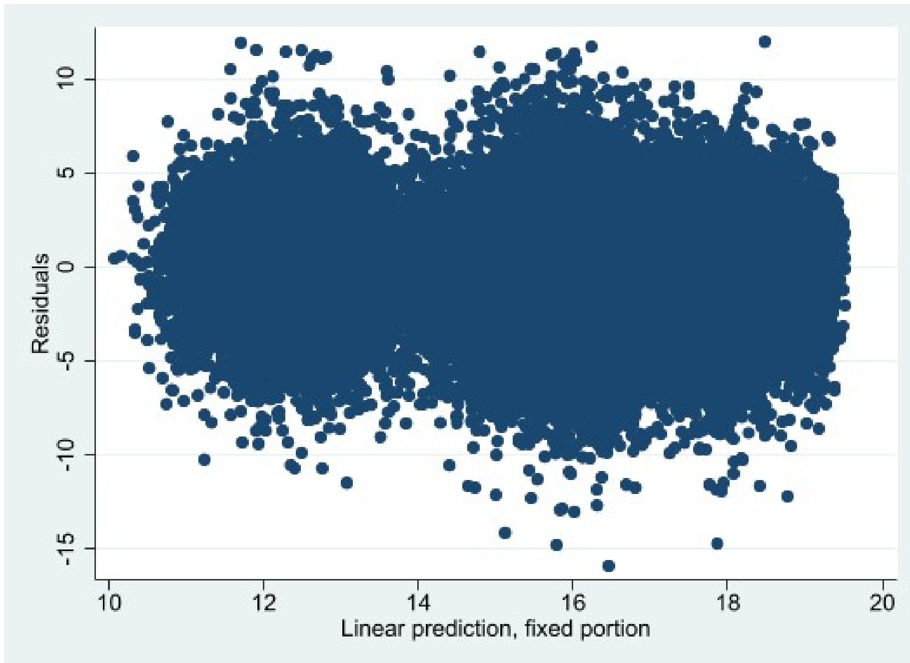


Figure 21: Model 6 Residual vs Fitted Plot

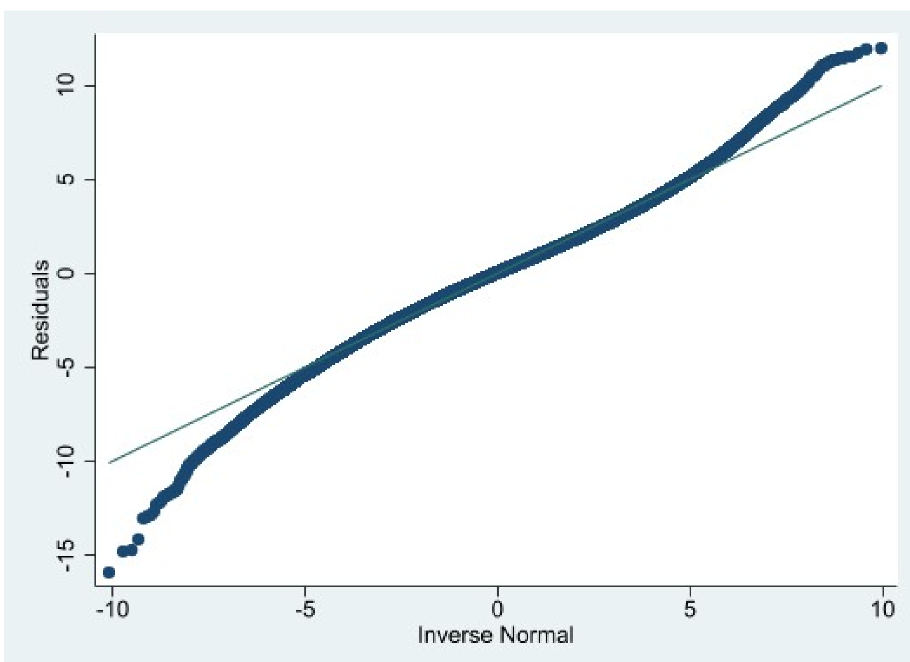


Figure 22: Model 6 Residual Quantile Plot

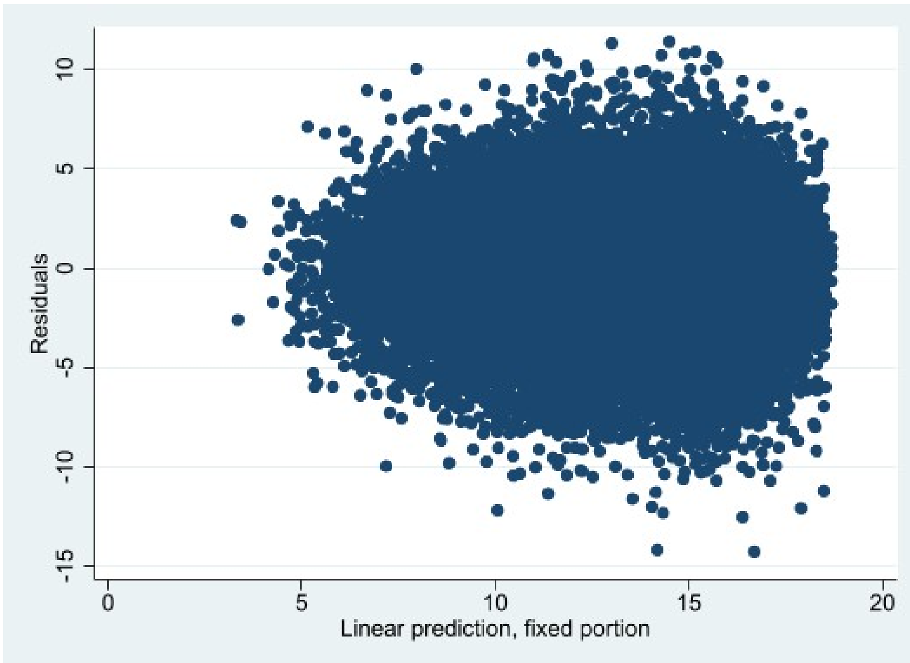


Figure 23: Model 7 Residual vs Fitted Plot

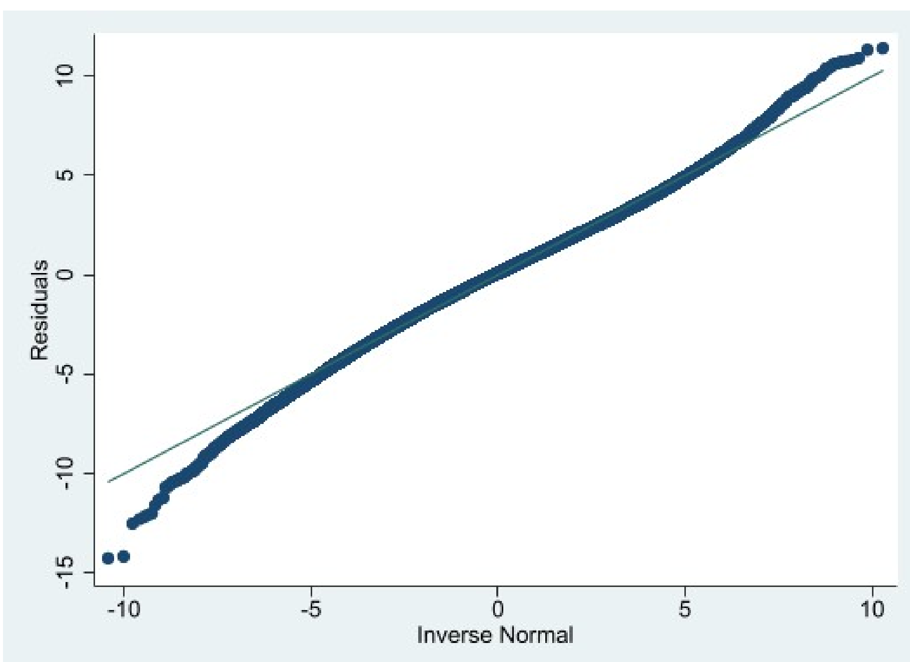


Figure 24: Model 7 Residual Quantile Plot